

PV–PWR power board

Rev A2 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PV–P75: 3 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

01 PC connector, command buffers, status

interfaces.PC (2x32) to the control board, 3,3 → 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points

File: sheets/01_pc.kicad_sch

02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/02_ph1_power.kicad_sch

03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03_ph1_gda.kicad_sch

04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04_ph1_gdb.kicad_sch

05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/05_ph2_power.kicad_sch

06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06_ph2_gda.kicad_sch

07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07_ph2_gdb.kicad_sch

08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/08_ph3_power.kicad_sch

09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09_ph3_gda.kicad_sch

10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/~3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10_ph3_gdb.kicad_sch

11 Port A power path (pv)

Varistor Y + monitor

File: sheets/11_leanA.power.kicad_sch

12 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/12_leanA.ctrl.kicad_sch

13 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/13_leanA.drive.kicad_sch

13 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/13_leanB.power.kicad_sch

14 Port B sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/14_leanB.ctrl.kicad_sch

15 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/15_leanB.drive.kicad_sch

15 Terminals, EMI filter, insulation test

CAD CTRNOL is non-regulation Quesing error! Xsserisavefor number on each build (output PV–PWR design checked kEXGDAUVR38822)rev 3 TronSG2430 form, is SW-G ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651, T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX75 from force transformer (AUXJ-BE-LV).CC output 100%). Shared heatsink: Tj 107 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 19.8 W (economiser) <= aux live rating 30.0 W; module aux 62 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/15_terminals_jmd.kicad_sch

16 Live 5 V / 3.3 V, SELV connector

Live 23.62–24.68 V (FB); SELV 21.4–30.1 V over the load table, gates off 21.4–27.7 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low

File: sheets/16_supplies.kicad_sch

17 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–; 2 x 20R R pulse-rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 35LHV (AUX75T film forced windings (AUXJ-BE-LV).CC output 100%).

File: sheets/17_aux75_input.kicad_sch

18 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms

File: sheets/18_aux75_startup.kicad_sch

19 AUX75 PWM, SiC, clamp, AUX–T1

UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 50:9:10; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/19_pwm.kicad_sch

20 AUX75 live 24 V, SELV 24 V, status

Live 23.62–24.68 V (FB); SELV 21.4–30.1 V over the load table, gates off 21.4–27.7 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low

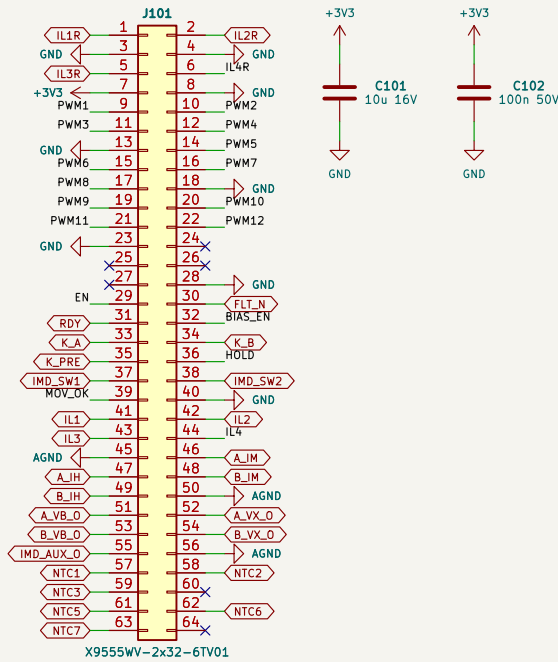
File: sheets/20_status.kicad_sch

Sheet: /
File: PV–PWR.kicad_sch
Title: PV–PWR power board – Cover / Hierarchy

Size: A3	Date: 2026–10–05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 1/23	

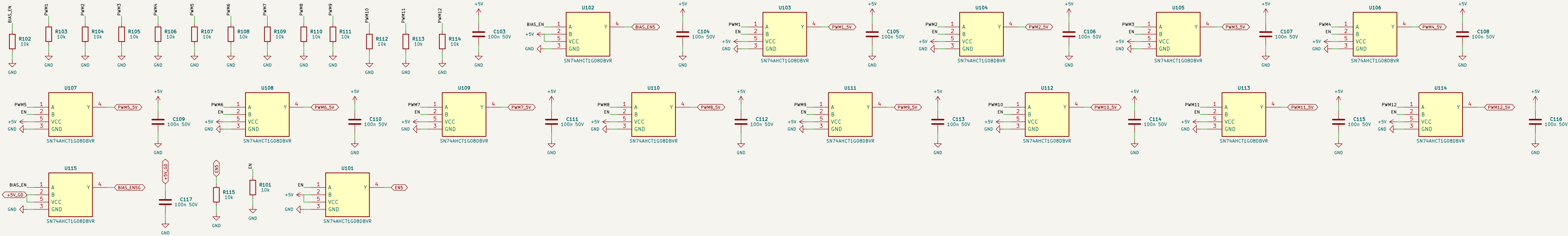
File: sheets (20 rows) - outputs: bind, ext

PC CONNECTOR (INTERFACES.PC), LIVE



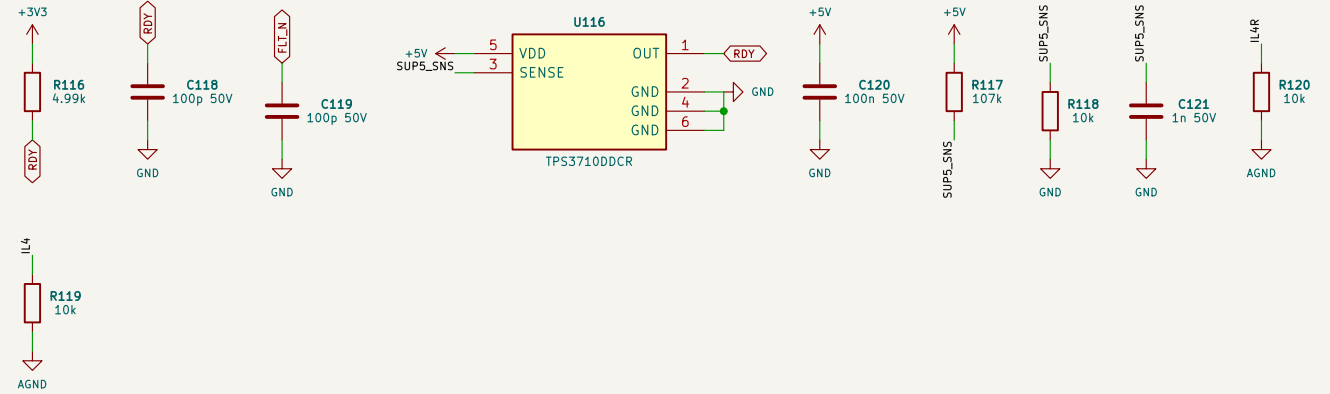
Both boards live: GND = AGND = BUS-. +3V3 goes TO the control board (rev A2; the former +24V / +5V pins carry the sensor references IL1R-IL4R). Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).
PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 and IL4R pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



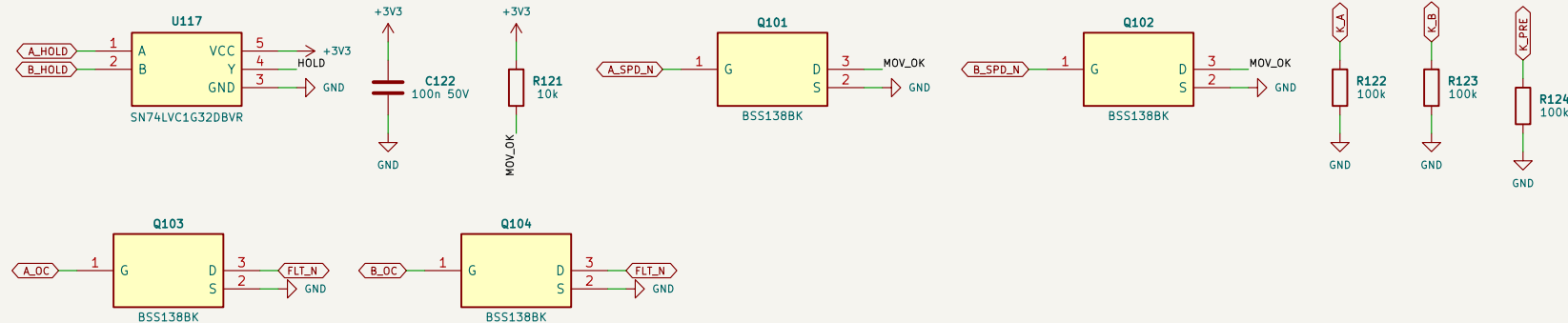
NSI6651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



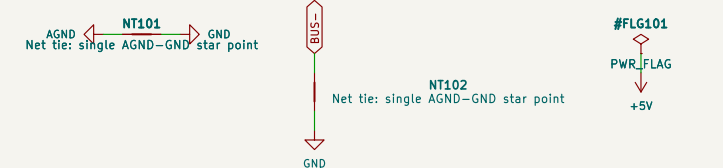
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
+5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHC2 VCC <= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



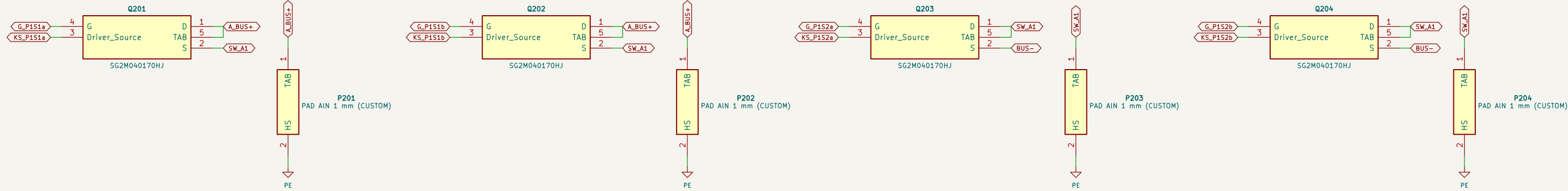
HOLD = A_HOLD OR B_HOLD ([Lport] >= 0.97-1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



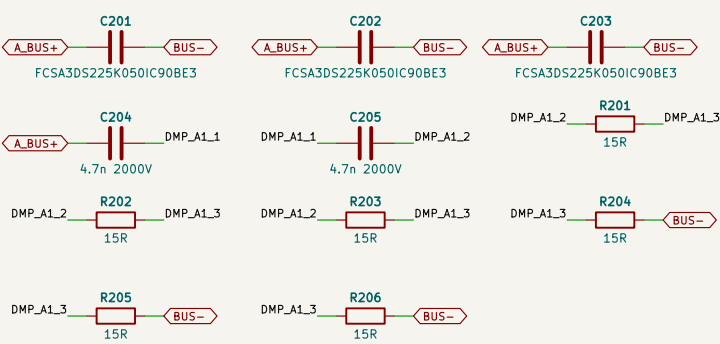
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-); Layout: gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



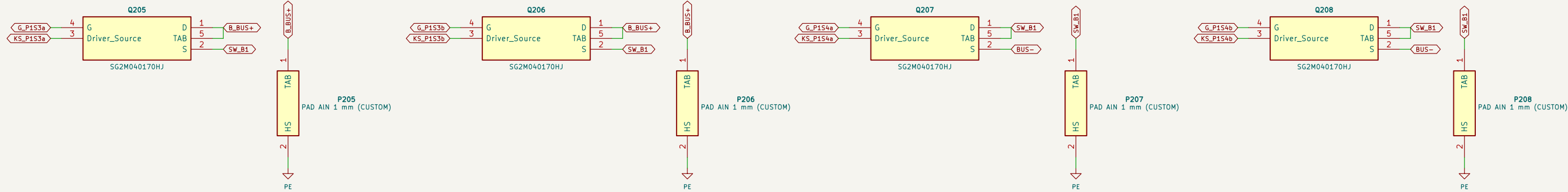
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



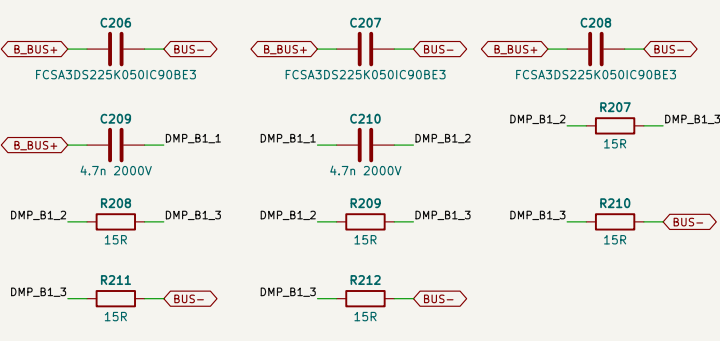
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



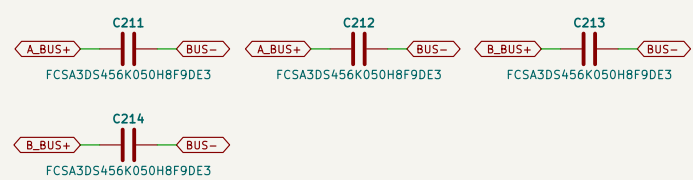
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
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PHASE 1 LEG B DECOUPLING + RC DAMPER



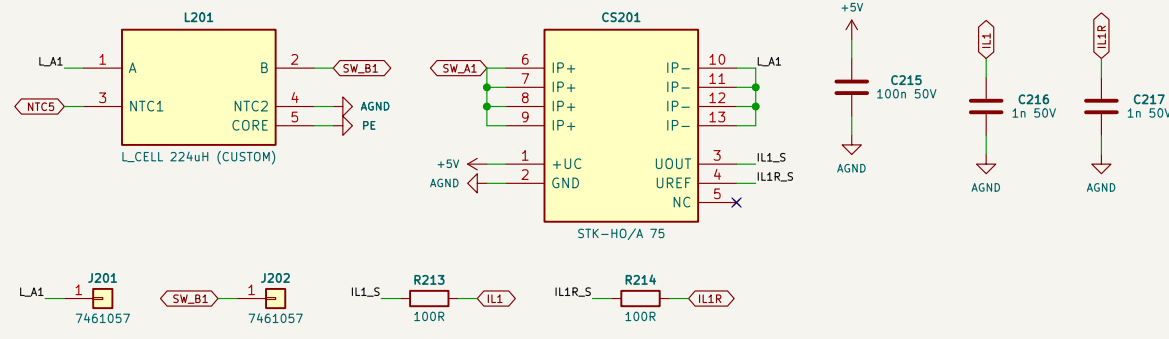
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



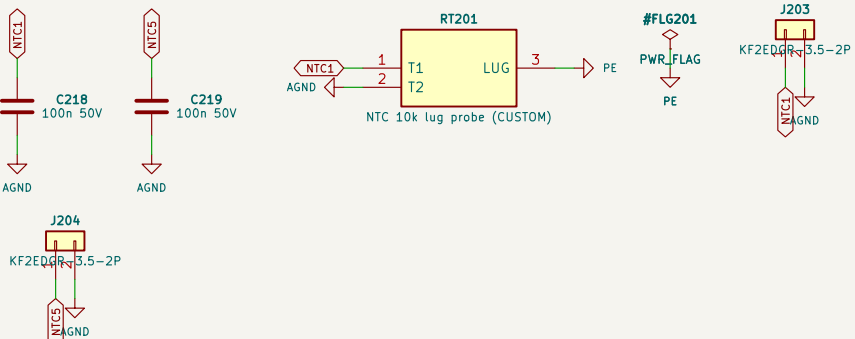
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



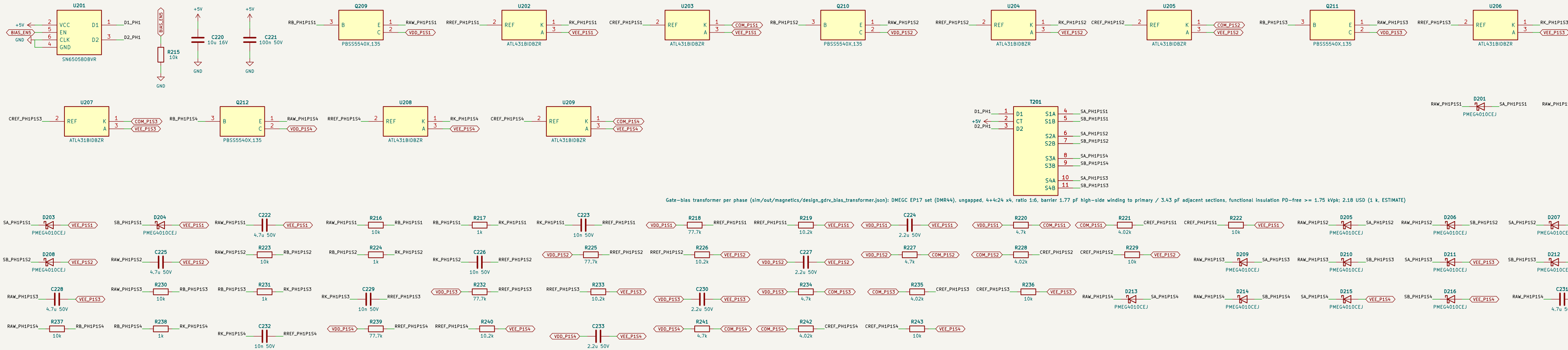
Lead A from SW_A1 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out + = SW_A -> SW_B) to stud
LA1: lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF: PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



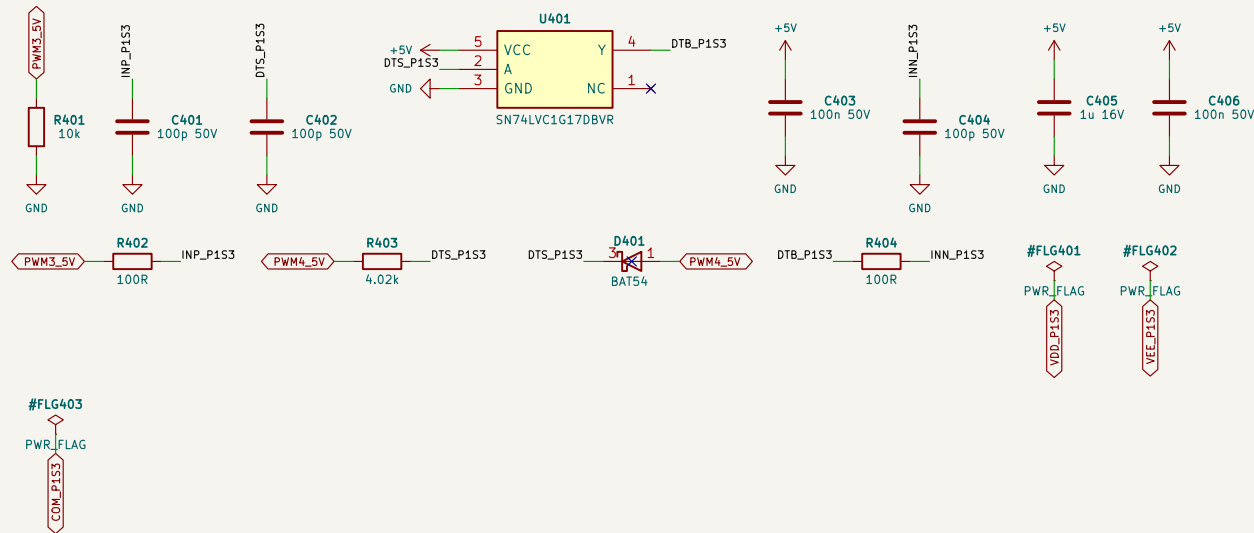
NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



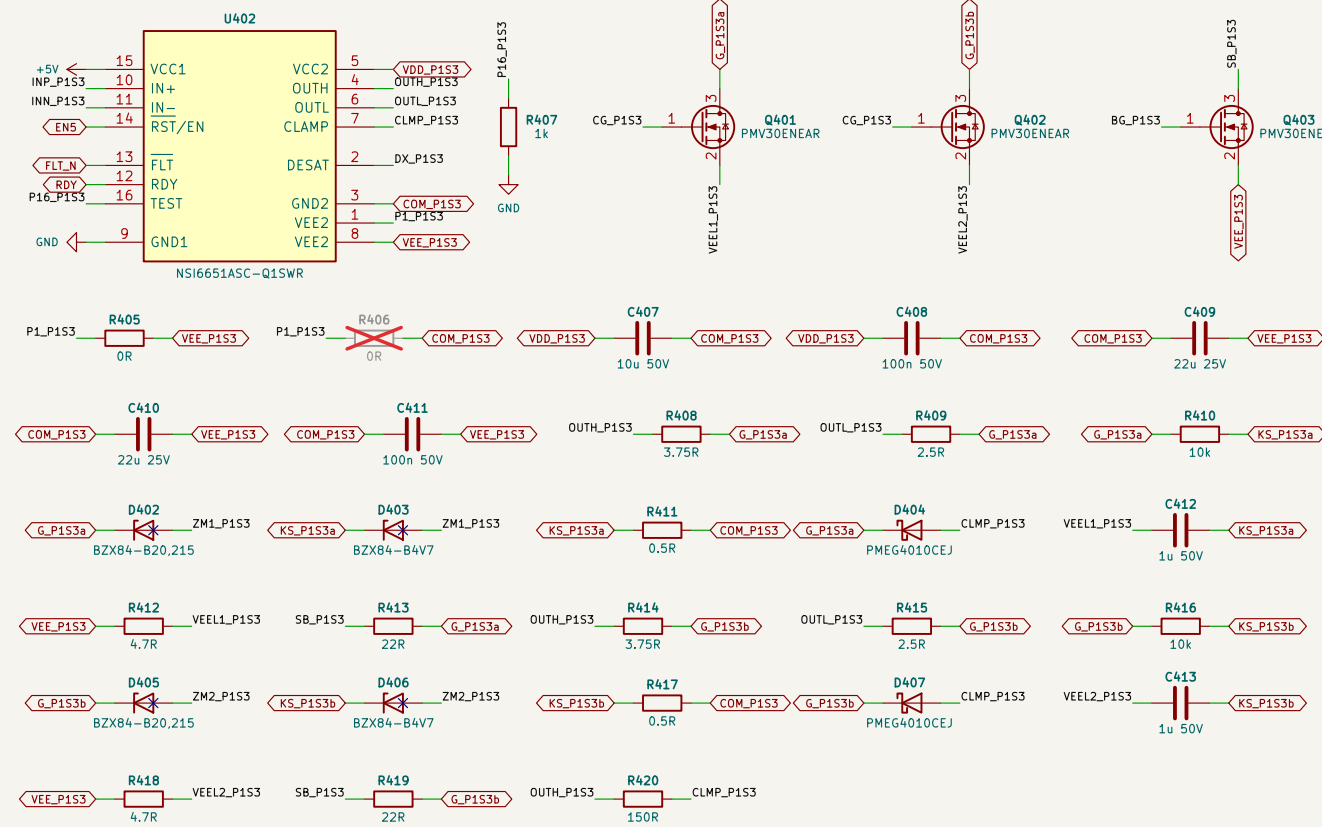
SN6505B push-pull 420 kHz from +5V. BIAS_EN high = on: T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S3: LOGIC SIDE



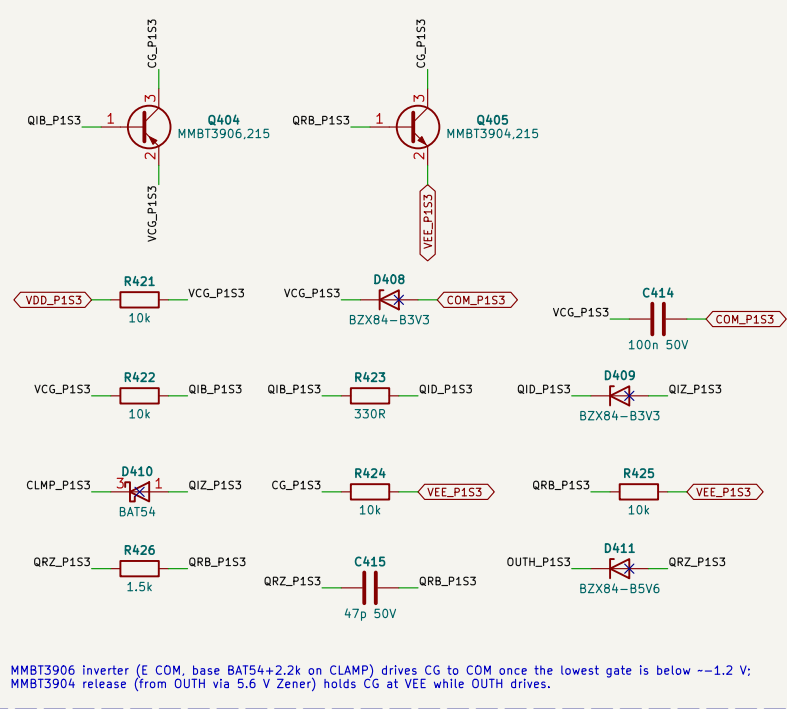
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



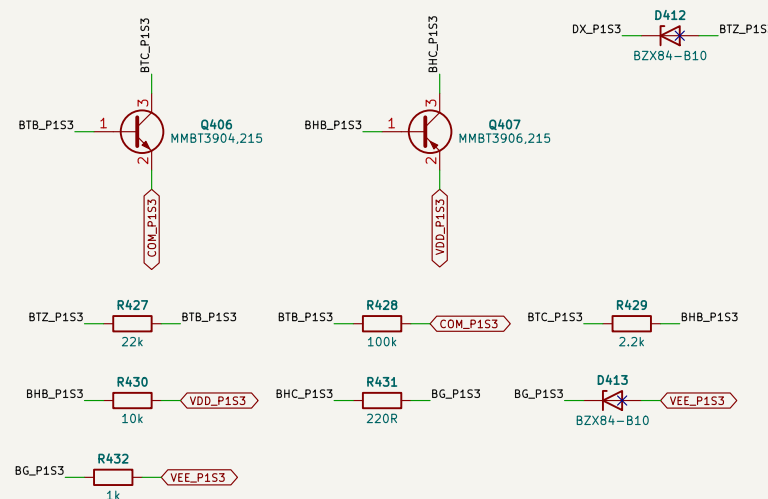
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30EENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



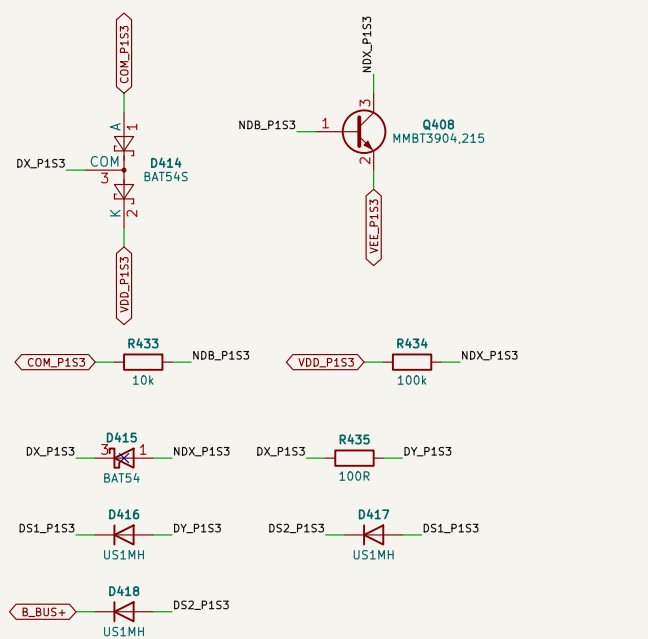
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



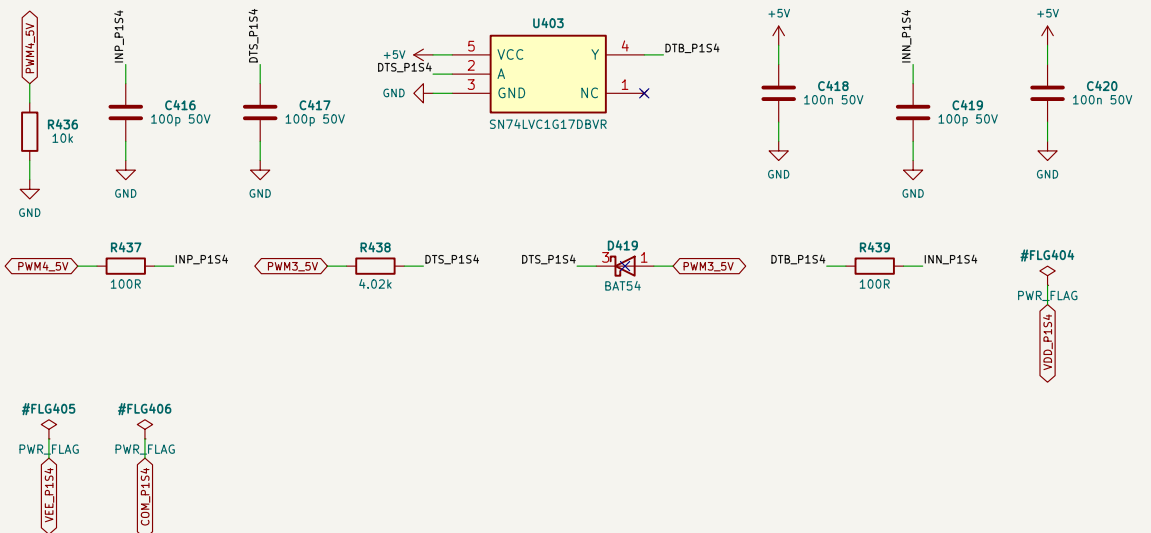
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P1S3: DESAT



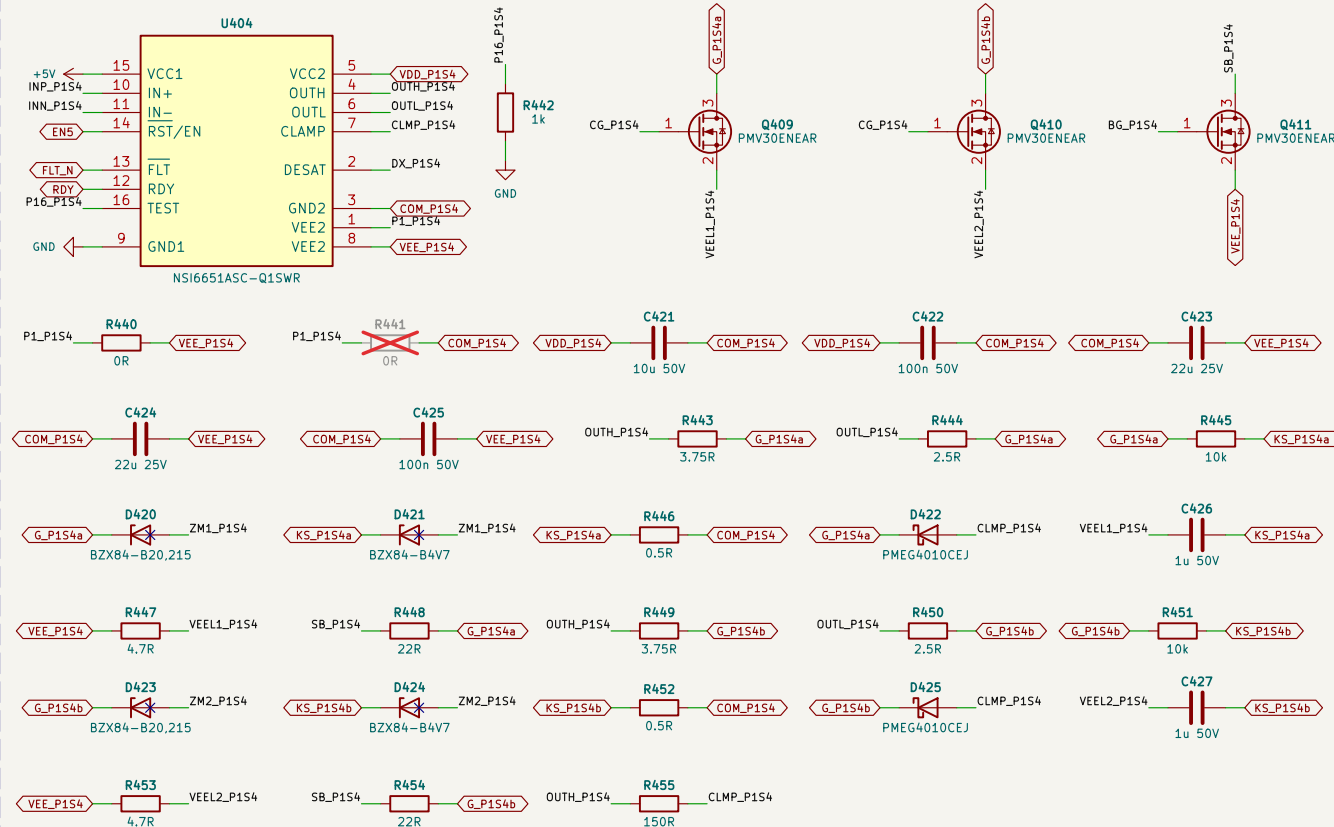
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S4: LOGIC SIDE



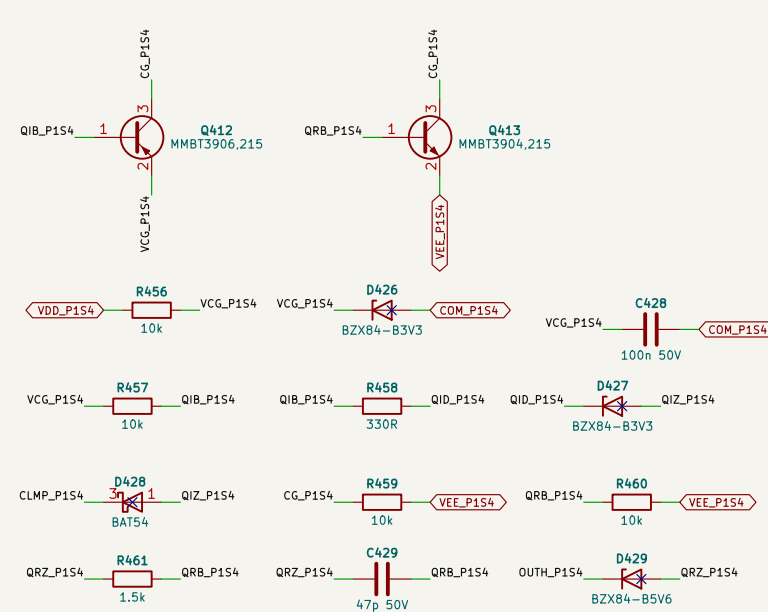
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



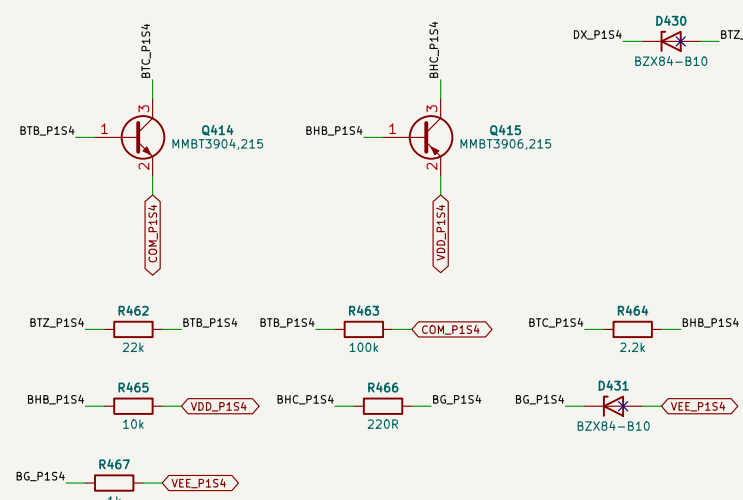
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
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P1S4: MILLER CLAMP CONTROL



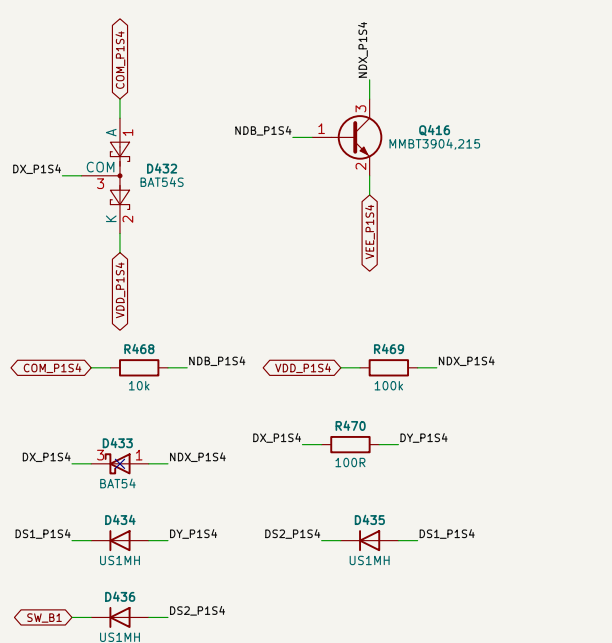
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MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P1S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 5

Schematics authored by Chinmoy Bhuyar
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

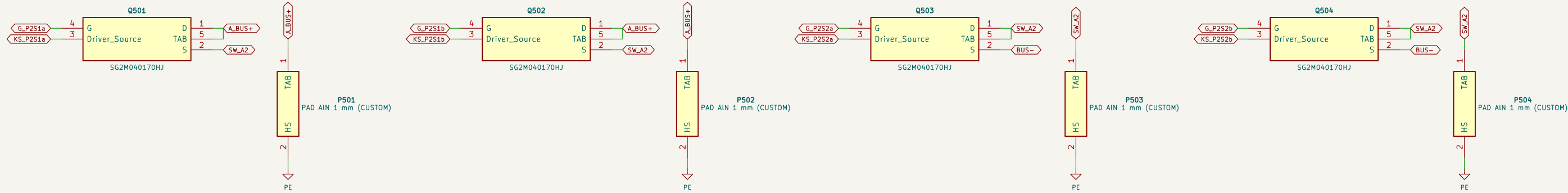
Sheet: /04 Phase 1 leg B gate drive (S3, S4)/
File: 04_ph1-gdb.kicad_sch

Title: PV-PWR power board - Phase 1 leg B gate drive (S3, S4)

Size: A2 Date: 2026-10-05 Rev: A2

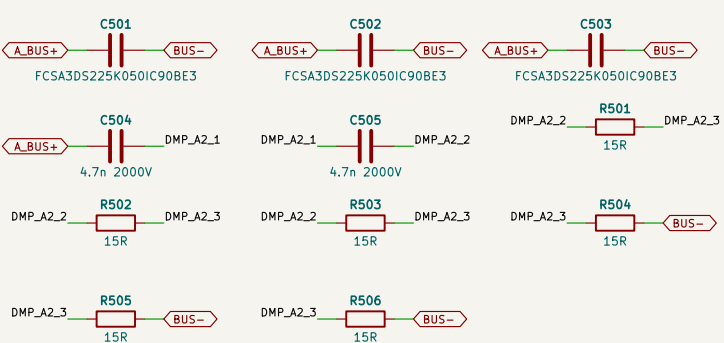
KiCad E.D.A. 10.0.6 Id: 5/23

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



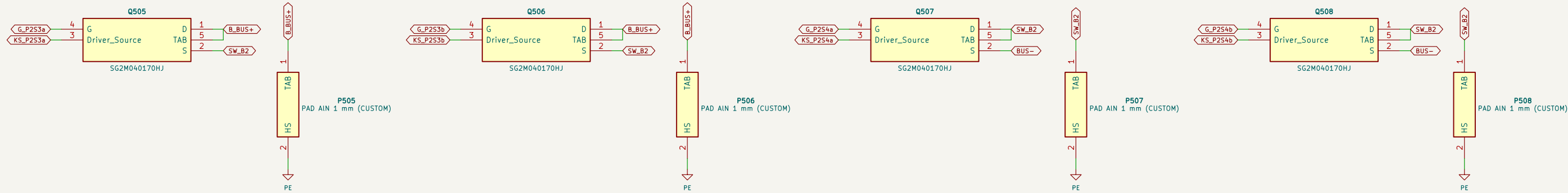
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



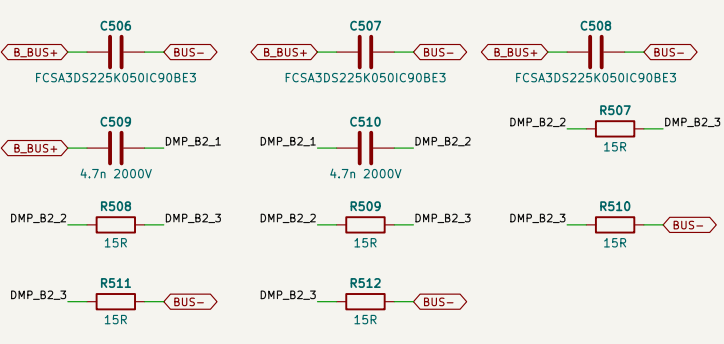
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



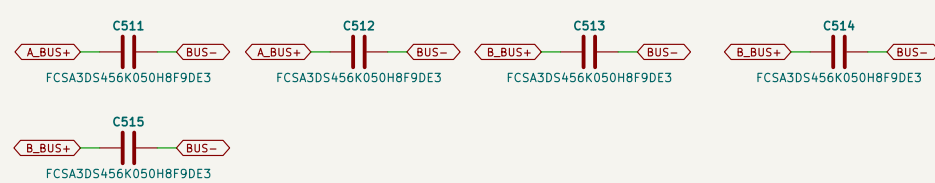
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



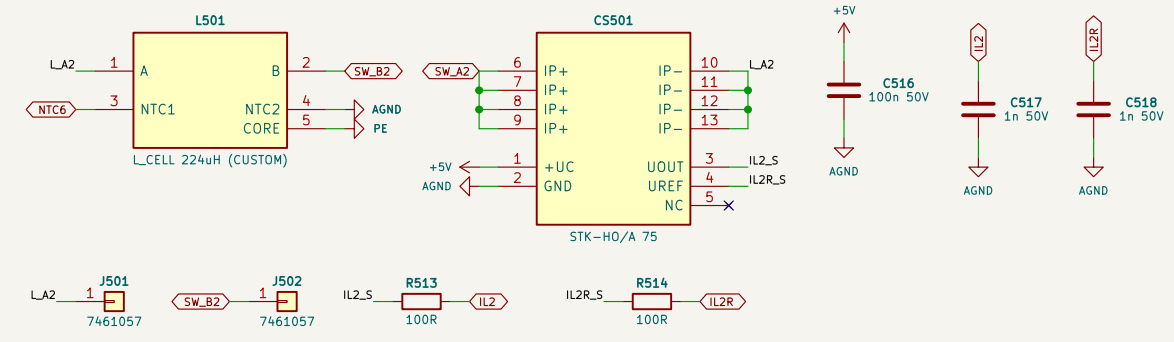
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PHASE 2 SHARE OF THE PORT BANKS



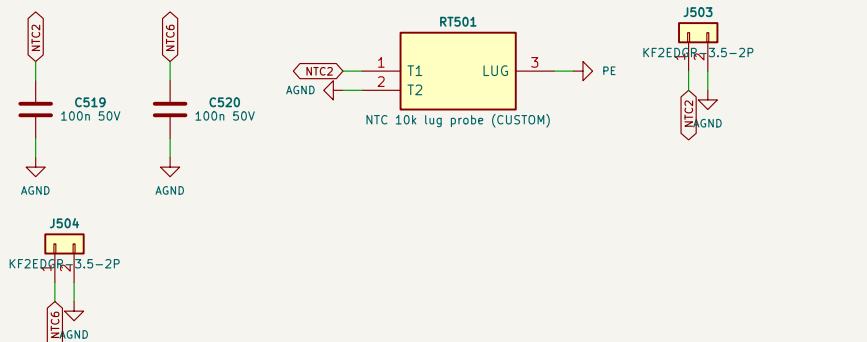
2 x FCSA3DS456 (45 uF) on A.BUS+ and 3 on B.BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF); the third on B.BUS+ is the battery side's seventh (D-056: full-power load rejection). Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



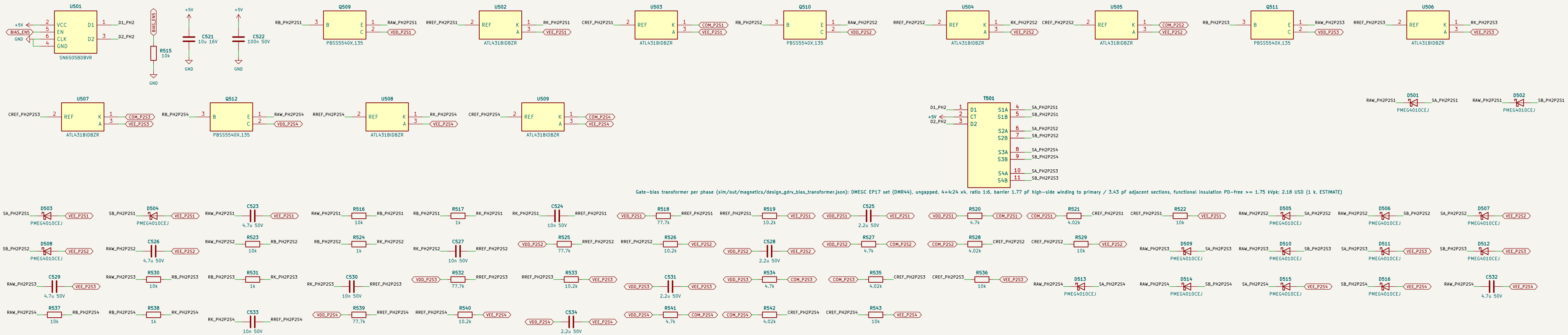
Lead A from SW_A2 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud LA2; lead B on stud SW_B2. IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100R = 1 nF at the source; IL2R = the sensor's own Uref (pin 4). 100R = 1 nF; PV-CT1 references its trip window to it. Primary at the switch node; functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



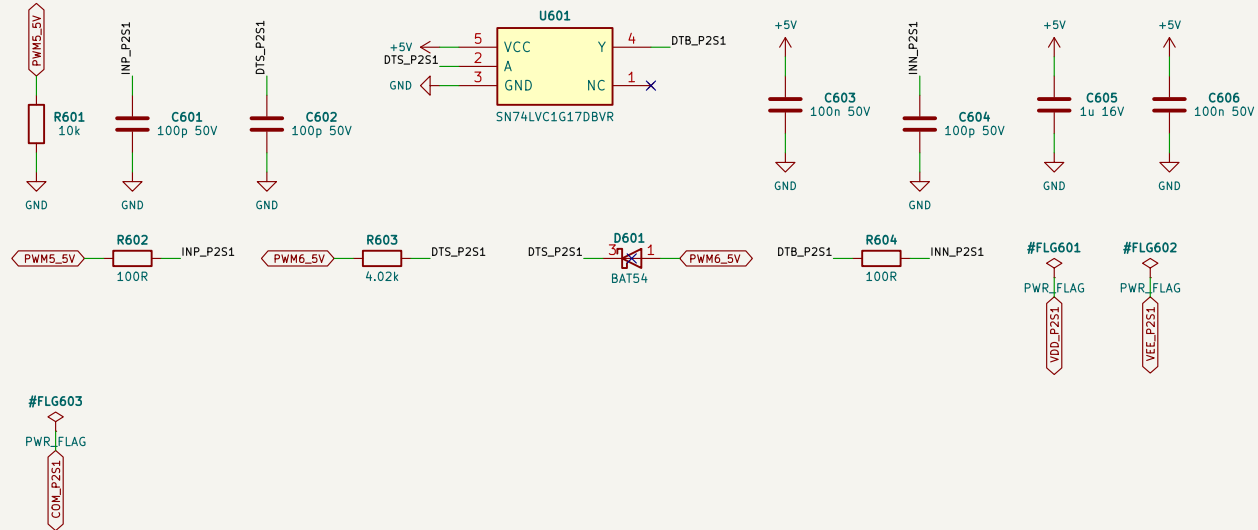
NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10). NTC6 = inductor winding NTC (functional to the winding). 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



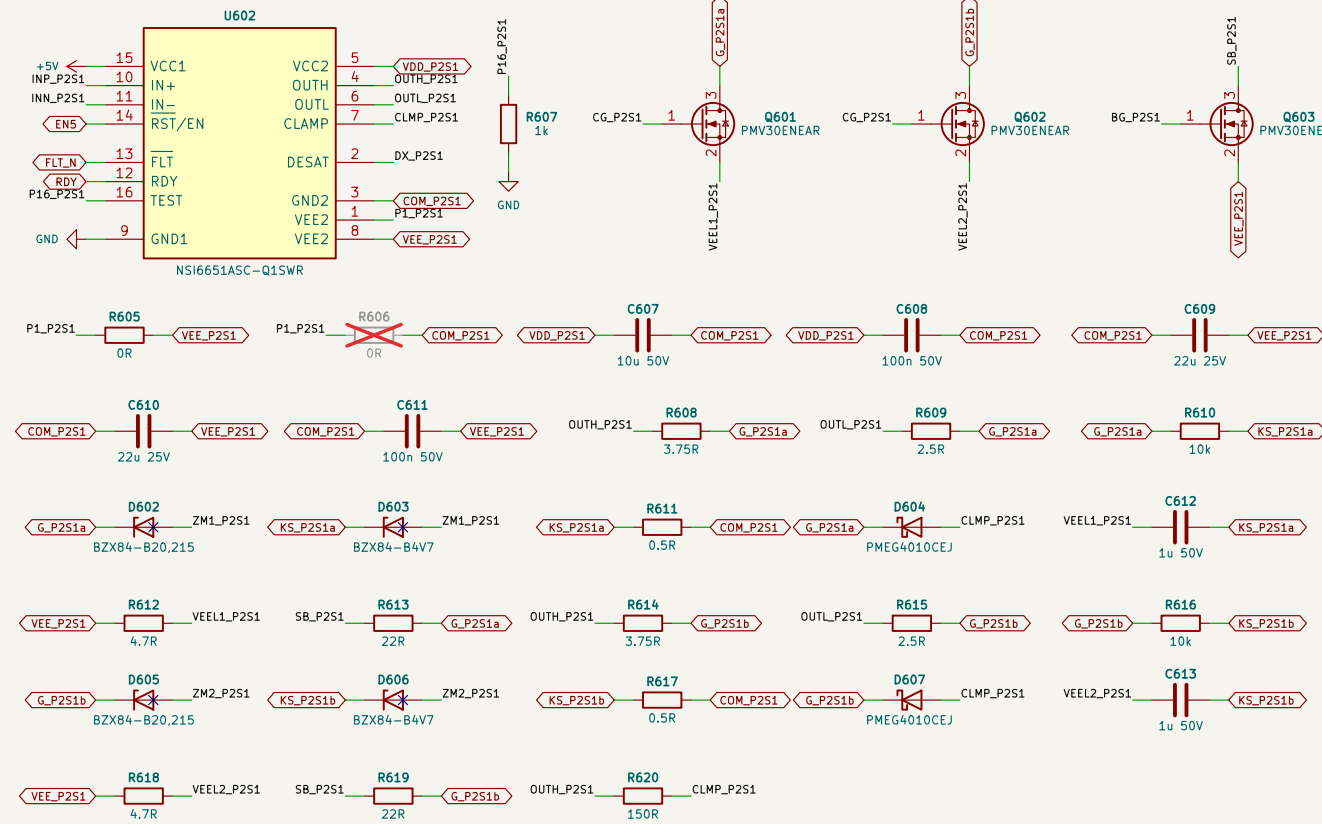
SN6505B push-pull 420 kHz from +5V. BIAS_EN high = on; T_BIAS4 custom; functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P2S1: LOGIC SIDE



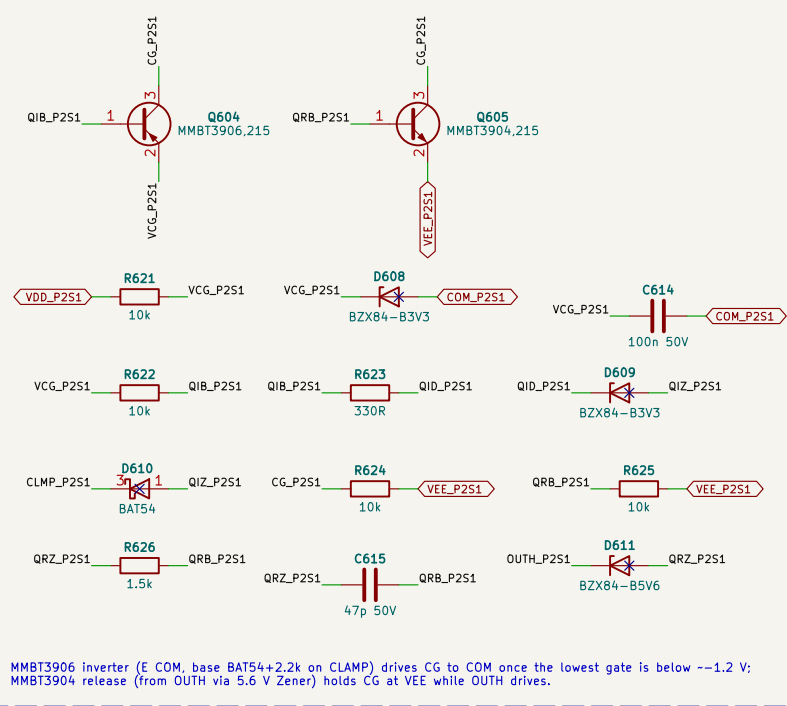
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



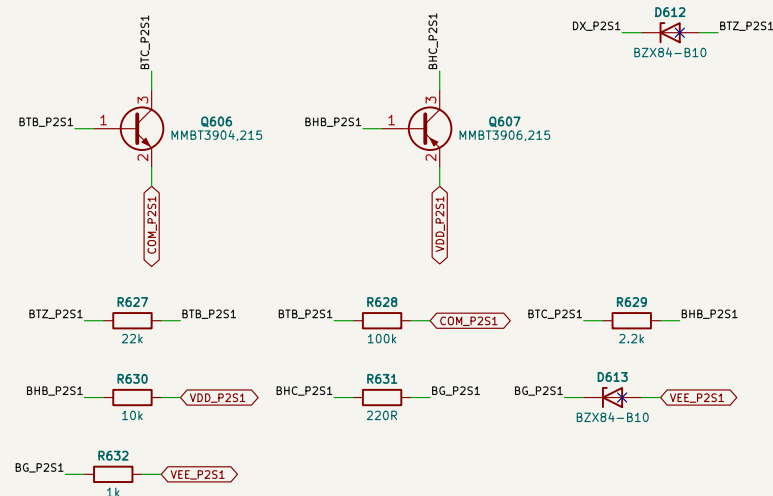
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMW30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL

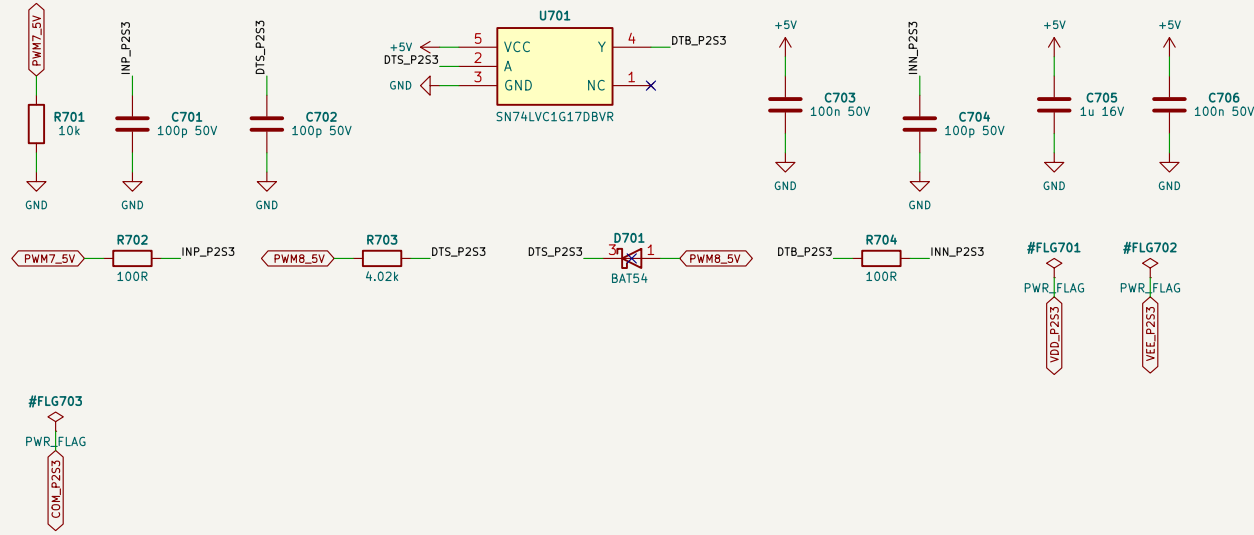


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER

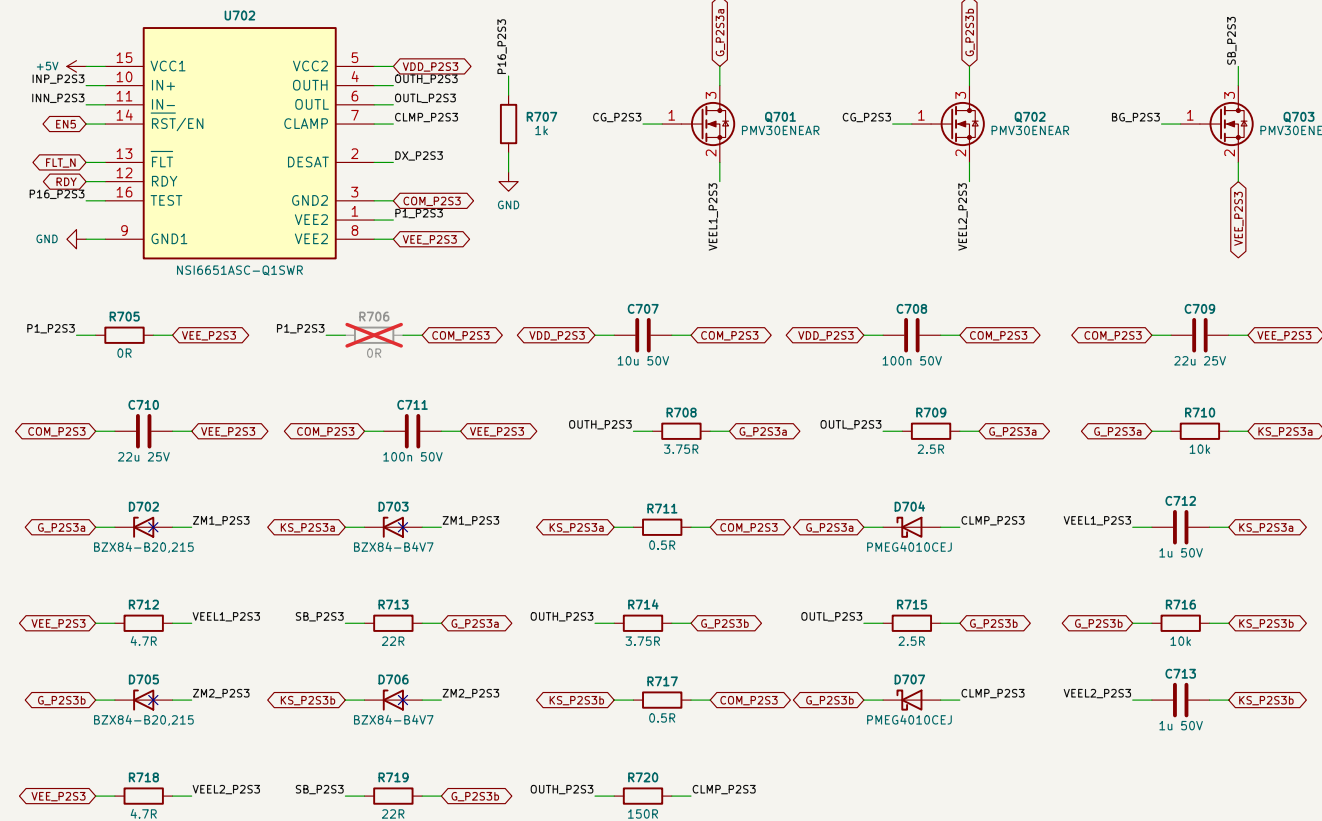


P2S3: LOGIC SIDE



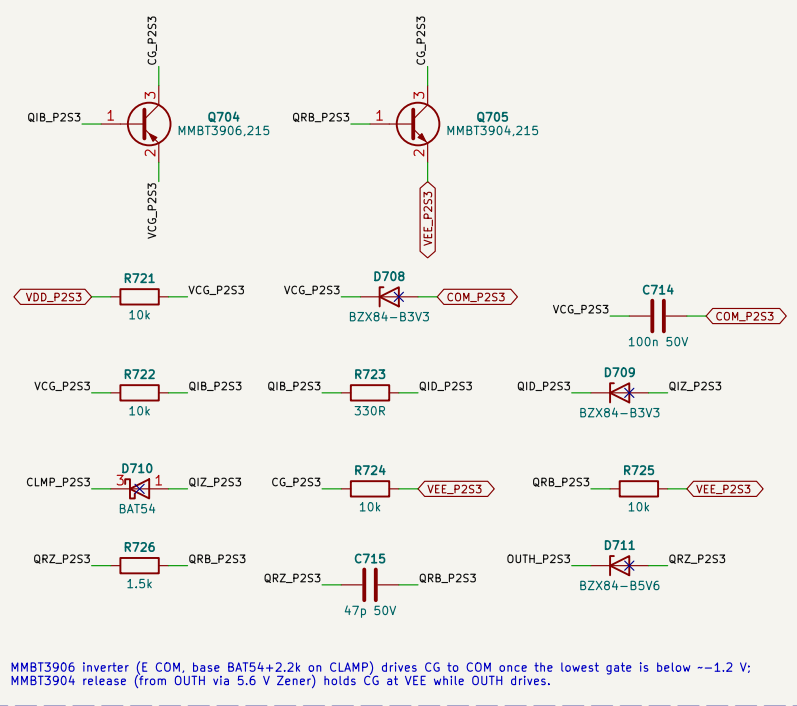
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



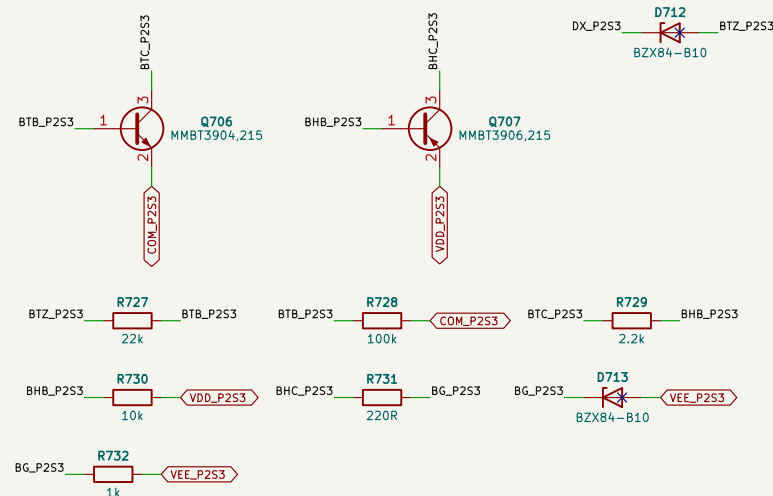
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL



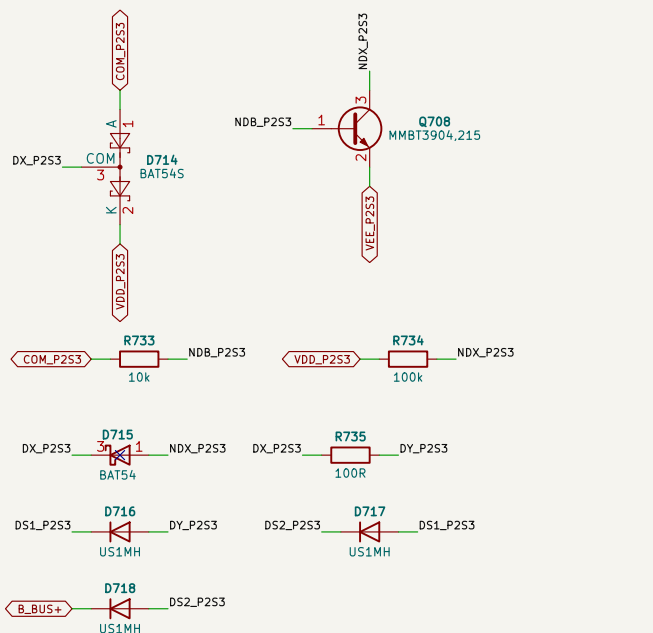
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER



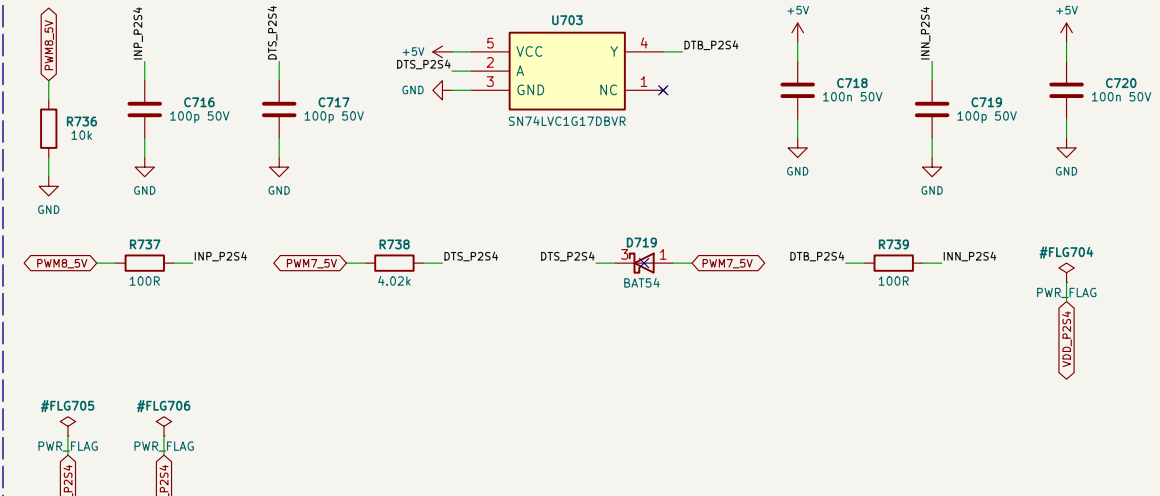
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P2S3: DESAT



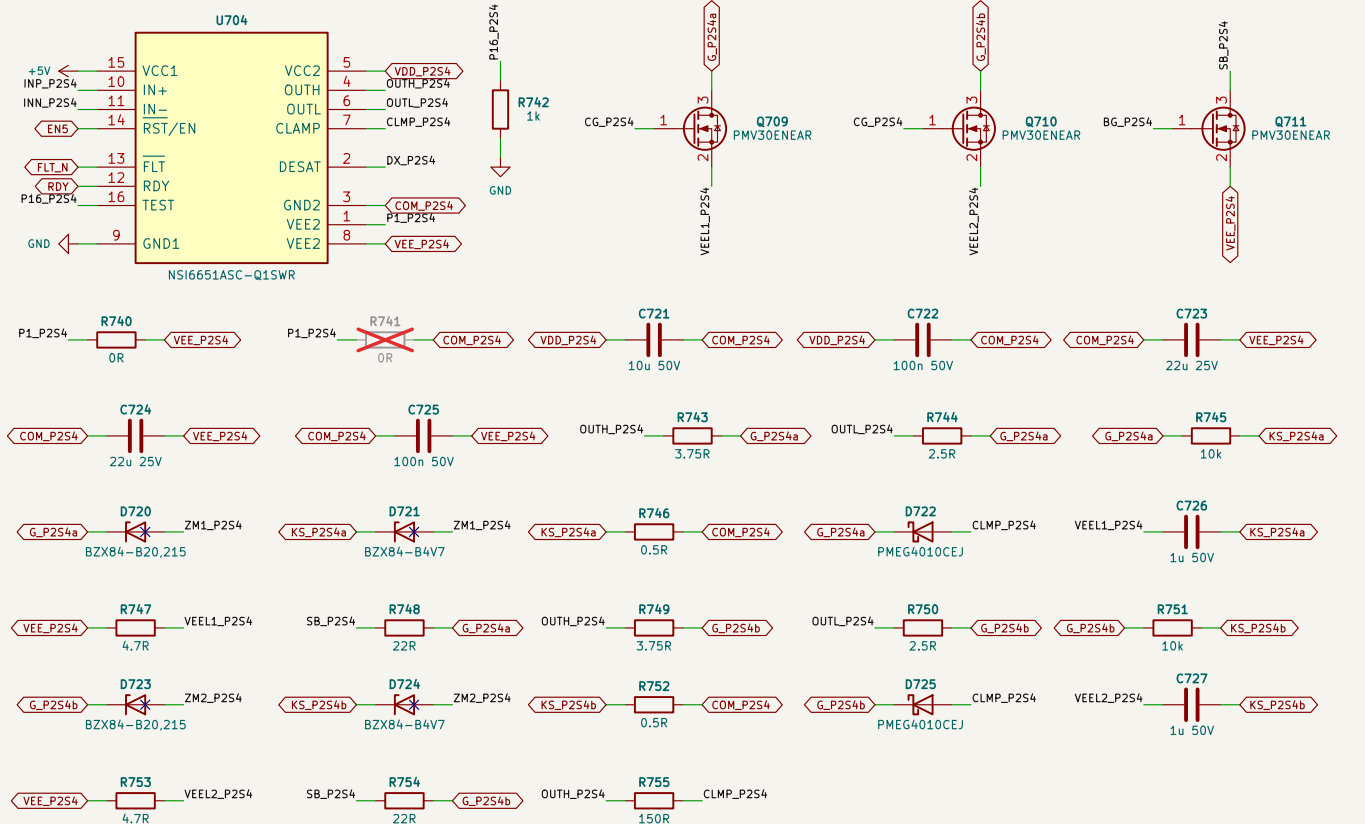
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S4: LOGIC SIDE



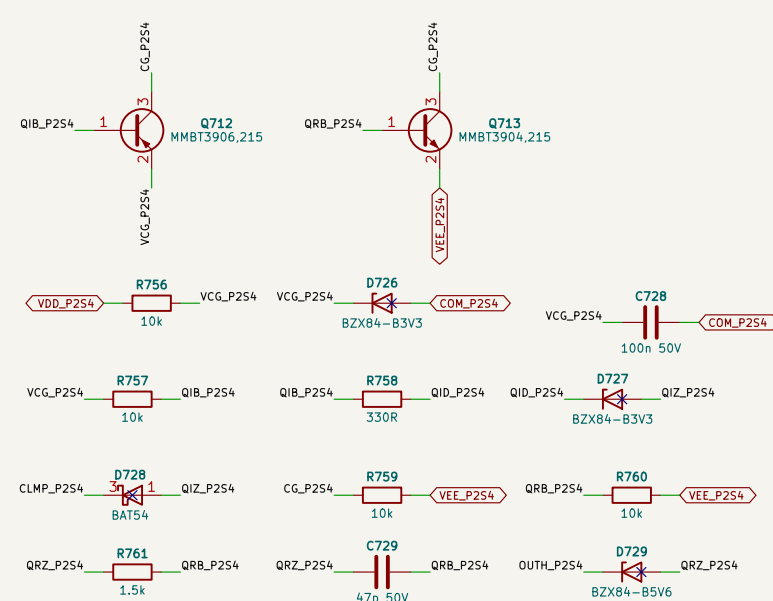
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S4: DRIVER AND 2 GATES



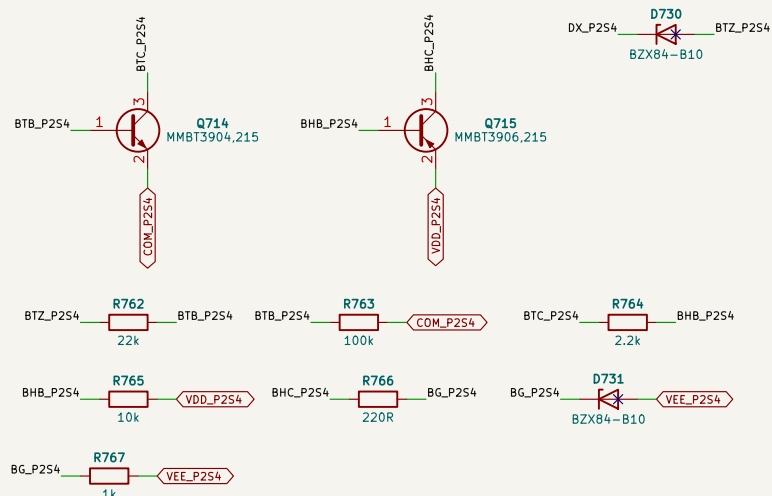
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S4: MILLER CLAMP CONTROL



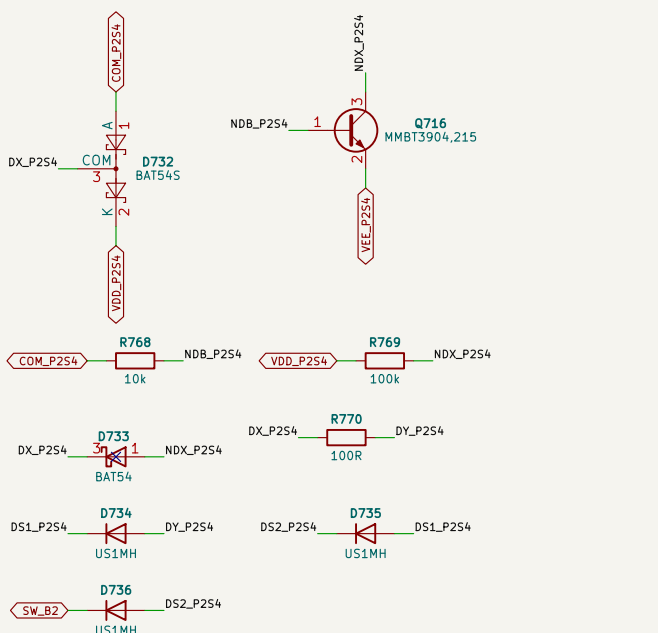
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P2S4: DESAT

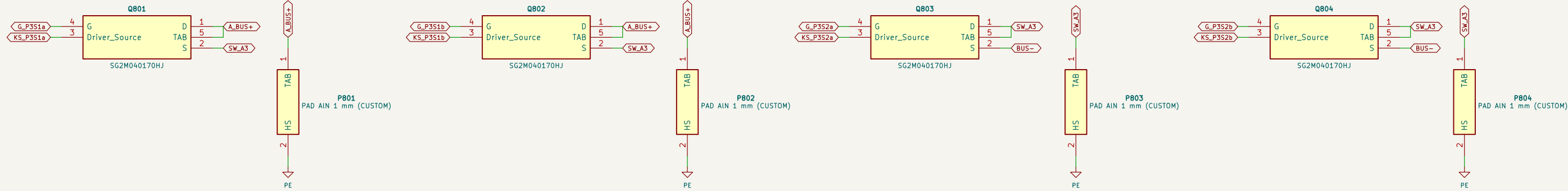


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

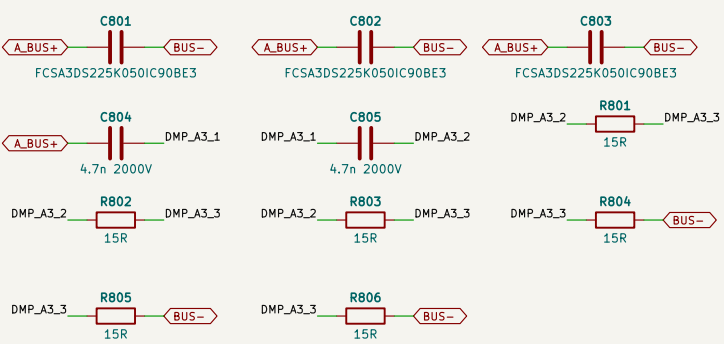
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_V at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nigspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



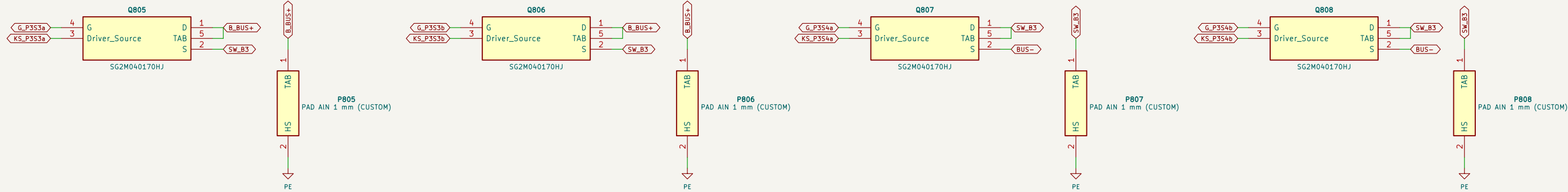
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



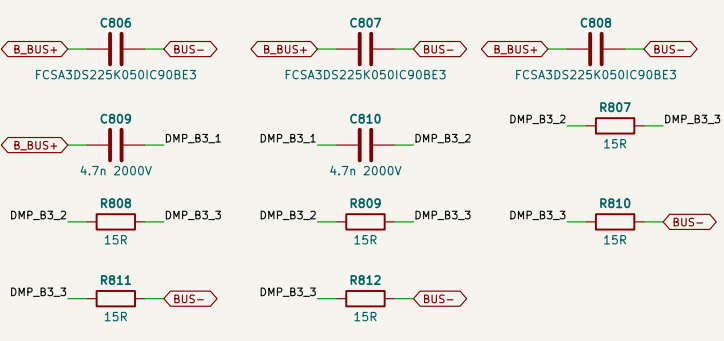
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



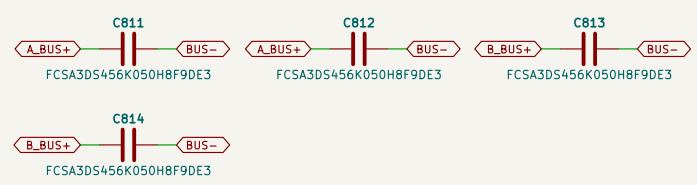
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



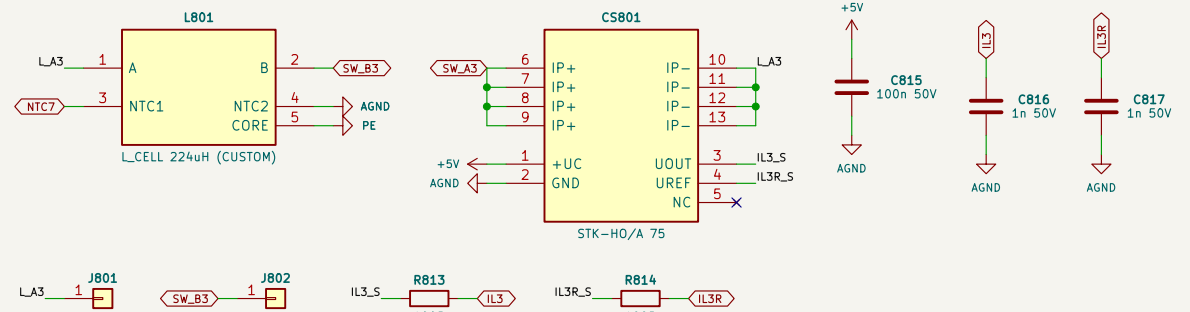
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



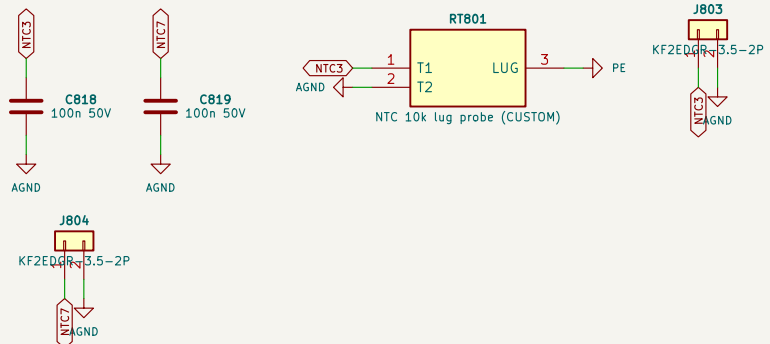
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



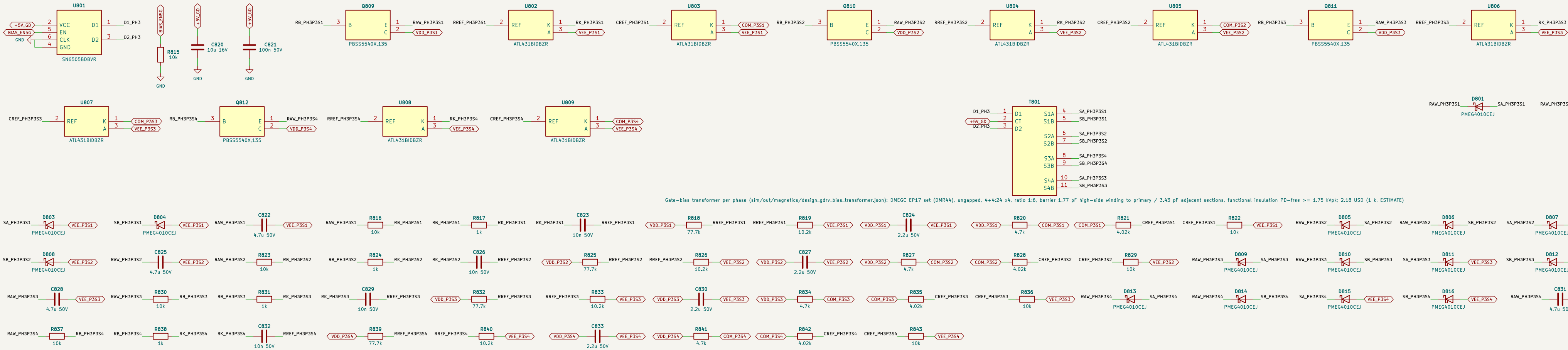
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud LA3; lead B on stud SW_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100R +/- 1 mV at the source. IL3R = the sensor's own Uref (pin 4). 100R +/- 1 mV: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC7 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T.BIAS4 custom, functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

[illegible]

P3S1: DRIVER AND 2 GATES

The schematic diagram illustrates the internal structure of the P3S1 driver and two PMV30GENAR gates (Q901 and Q902). The driver U902 is a 16-pin component with pins for VCC1, VCC2, IN+, IN-, RST/EN, FLT_N, RDY, P1_P3S1, GND1, GND2, and VEE1. It is connected to a 5V supply and a GND. The gates Q901 and Q902 are 3-pin components with pins for VDD_P3S1, VEE1_P3S1, and P1_P3S1. They are connected to a 5V supply and a GND. The diagram also shows various capacitors (C905, C906, C907, C908, C909, C910, C911, C912, C913, C914, C915, C916, C917, C918, C919, C920) and resistors (R905, R906, R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920) connected to the gates. The gates are connected to the P1_P3S1 and P2_P3S1 pins of the driver U902. The diagram is labeled 'P3S1: DRIVER AND 2 GATES'.

P3S2: LOGIC SIDE

P3S1: MILLER CLAMP CONTROL

The schematic diagram illustrates the Miller clamp control circuit for the P3S1. It features two inverters, Q904 (MMBT3906, 215) and Q905 (MMBT3904, 215), which are driven by the QIB_P3S1 and QRB_P3S1 signals, respectively. The outputs of these inverters are connected to a network of resistors (R921, R922, R923, R924, R925, R926), capacitors (C914, C915), and diodes (D908, D909, D910, D911) to control the CG_P3S1 and QRB_P3S1 nodes.

Key components and their values:

- Q904:** MMBT3906, 215 (Inverter)
- Q905:** MMBT3904, 215 (Inverter)
- R921:** 10k (Resistor)
- R922:** 10k (Resistor)
- R923:** 330R (Resistor)
- R924:** 10k (Resistor)
- R925:** 10k (Resistor)
- R926:** 1.5k (Resistor)
- C914:** 100n 50V (Capacitor)
- C915:** 47p 50V (Capacitor)
- D908:** BZX84-B3V3 (Diode)
- D909:** BZX84-B3V3 (Diode)
- D910:** BAT54 (Diode)
- D911:** BZX84-B5V6 (Diode)

The circuit is powered by VDD_P3S1 and VCG_P3S1. The output of the MMBT3906 inverter (Q904) drives CG_P3S1, which is connected to COM_P3S1 through a network of resistors and capacitors. The output of the MMBT3904 inverter (Q905) drives QRB_P3S1, which is connected to VEE_P3S1 through a network of resistors and capacitors.

MMBT3906 Inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V . MMBT3904 release (from QUTH via 5.6 V Zener) holds CG at VEE while QUTH drives.

P351: SHORT-CIRCUIT BOOSTER

[illegible][illegible][illegible]

channel length (+/-10 %) from the COM star: Ron, Roff, Kelvin
 device's gate-Kelvin pin: loop == 1 mV, DESAT string
 C=22, Miller hold (ngspice, gdrv); die +1.41 V vs VG5(h)
 and bias primary decoupled to GND locally (DM current).

[illegible]

3P52Z SHORT-CIRCUIT BOOSTER

Q914 MMBT3904

BTB_P352 1 2 3 BTC_P352

COM_P352

R962 22k BTB_P352 BTB_P352

R967 10k BHC_P352

R966 220V BQ_P352

R967 10k BQ_P352

VEE_P352

Q915 MMBT3906

BHB_P352 1 2 3 BHC_P352

COM_P352

R964 22k BTB_P352 BHB_P352

R965 10k BHC_P352

R966 220V BQ_P352

R965 10k BQ_P352

VEE_P352

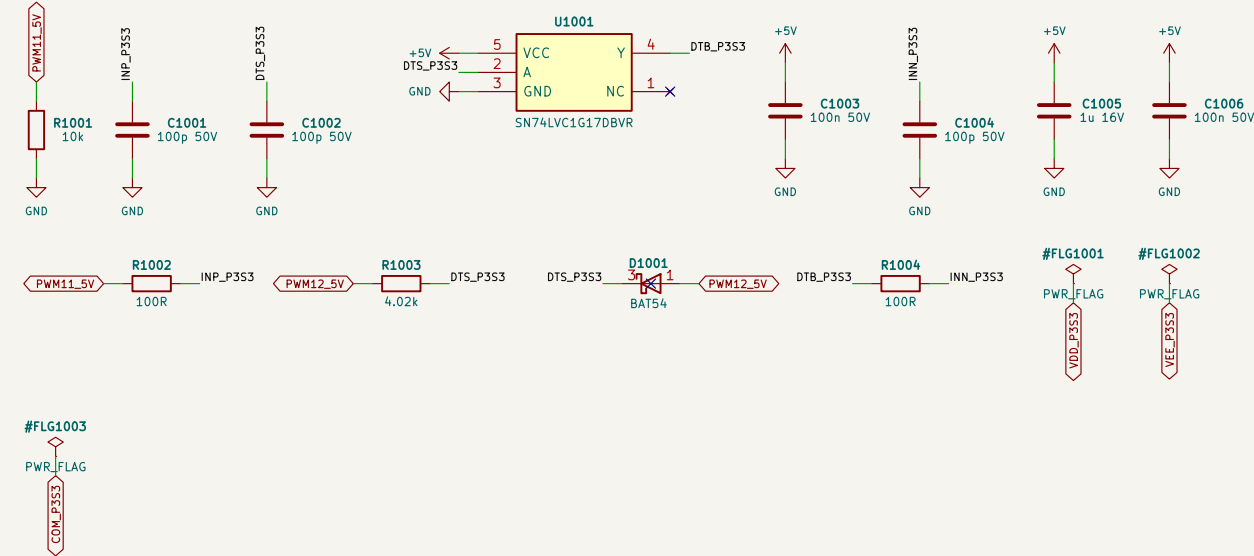
DESAT > 10.3–10.9 V (above the 10.0 V maximum trip of the –Q1 grade) → MMBT3904 → MMBT3906 charges BG to VEE + 10 V per device AG3400A + 22R pull the gate to VEE; BG holds ~3 vs (A = FET gate).

[illegible]

LEG A: LAYOUT RULES

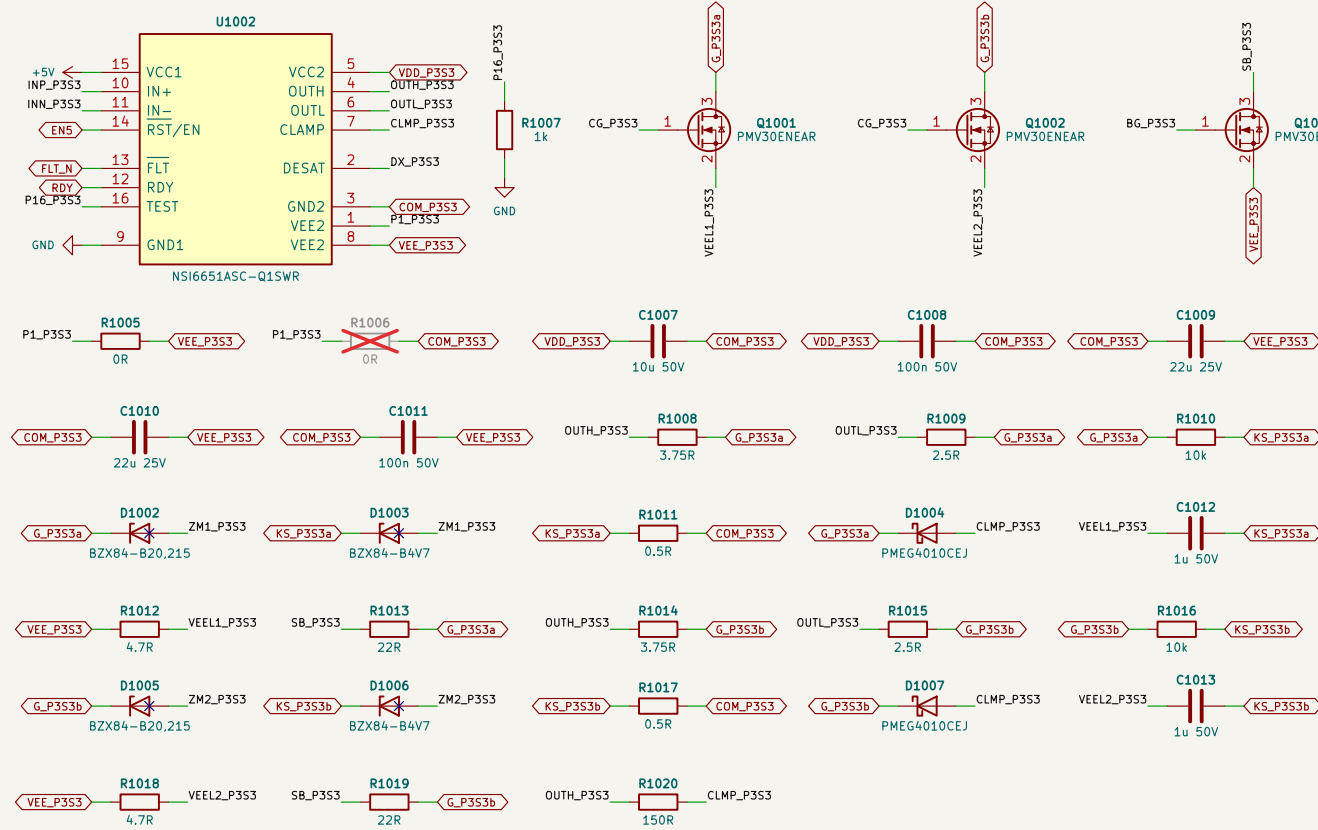
10	11
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P3S3: LOGIC SIDE



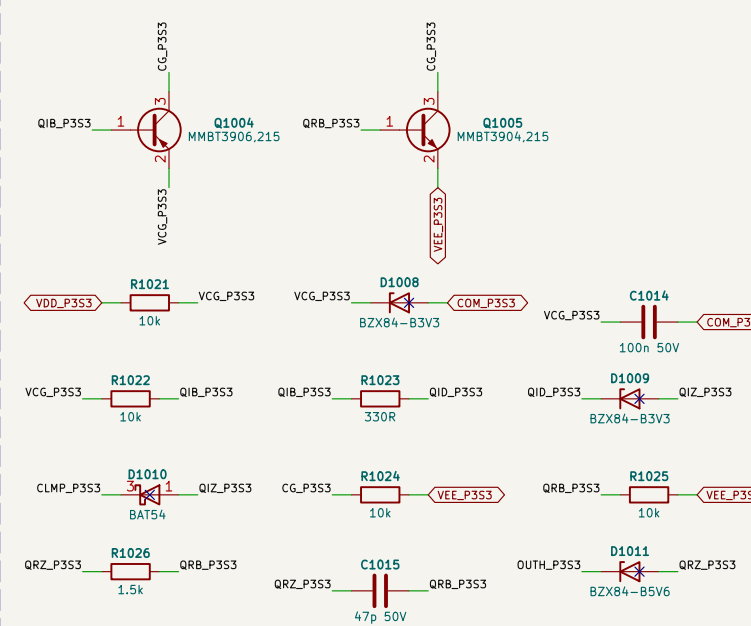
In+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



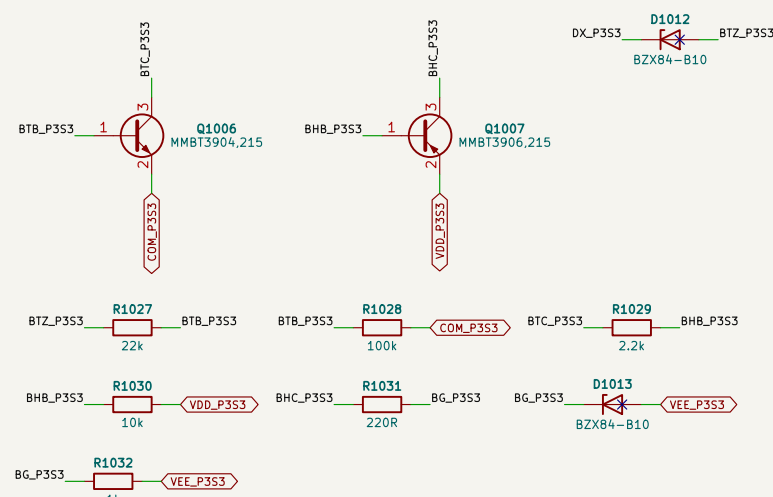
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



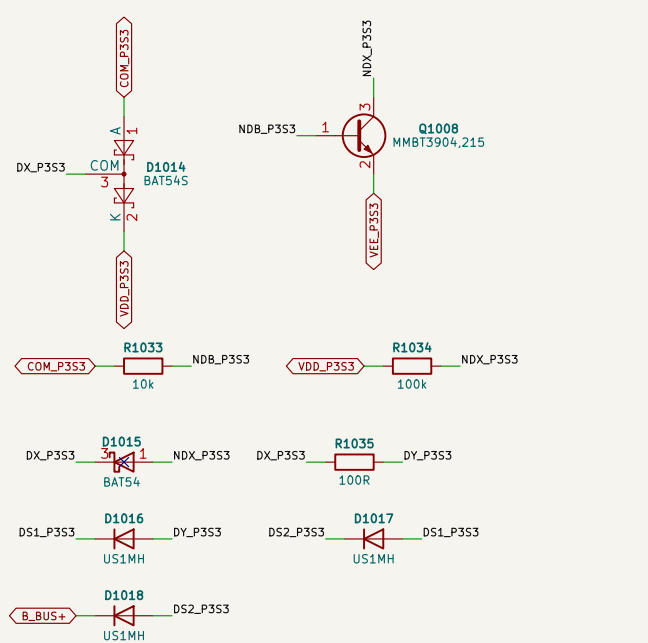
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



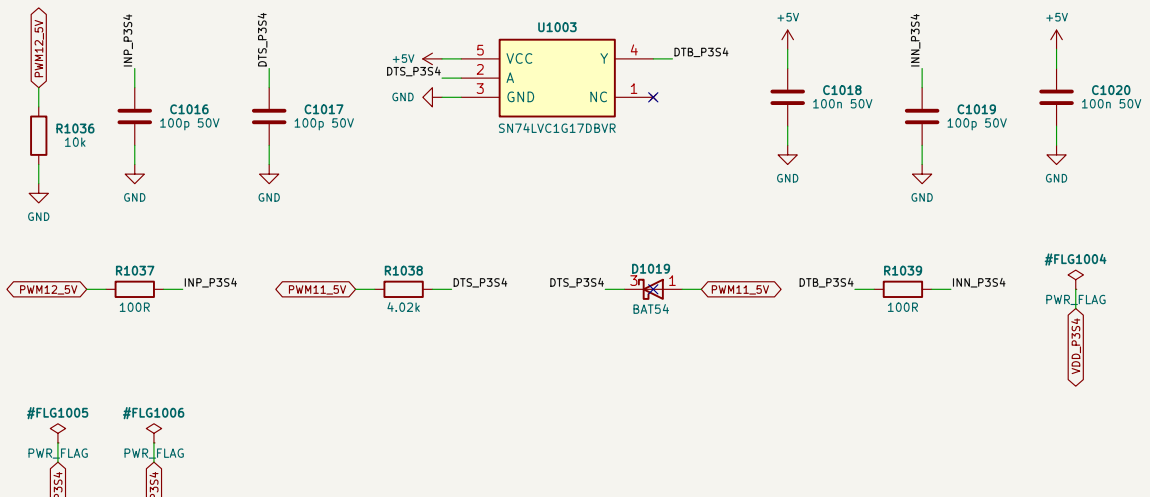
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S3: DESAT



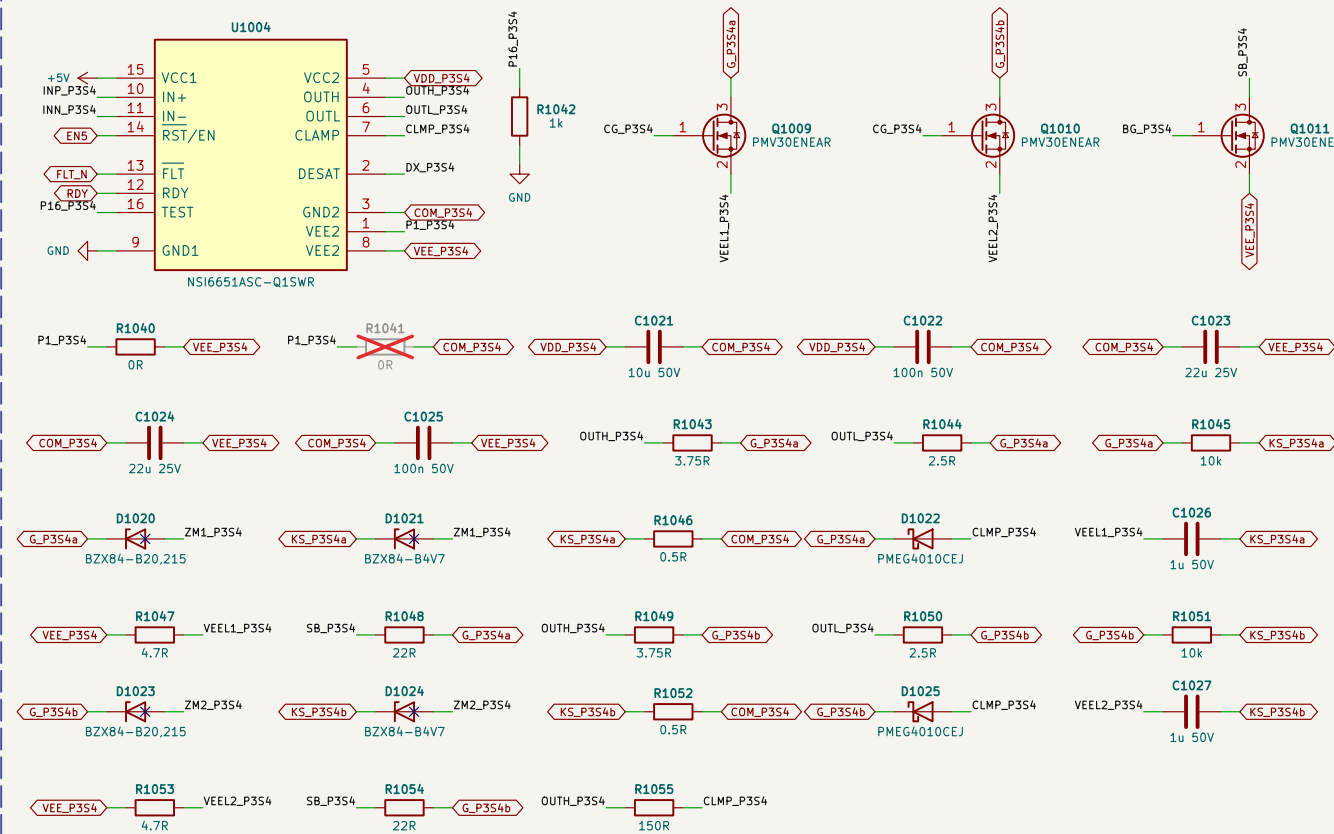
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S4: LOGIC SIDE



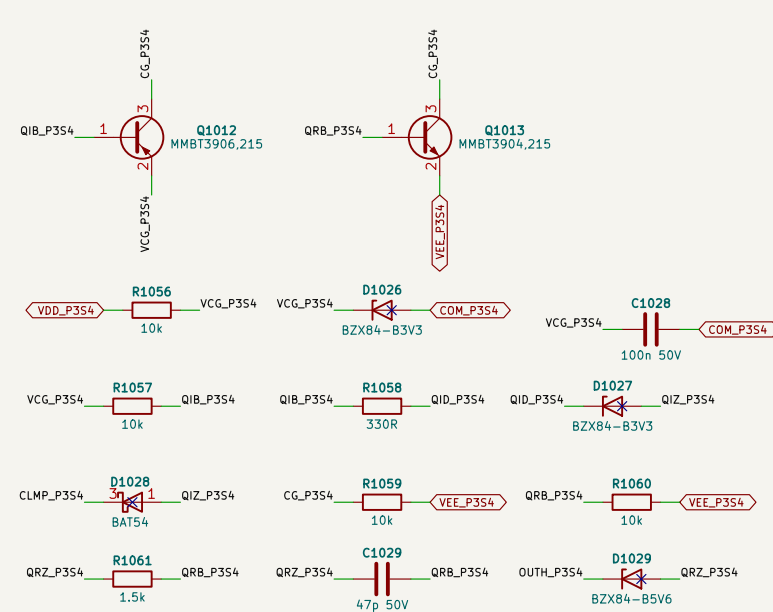
In+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



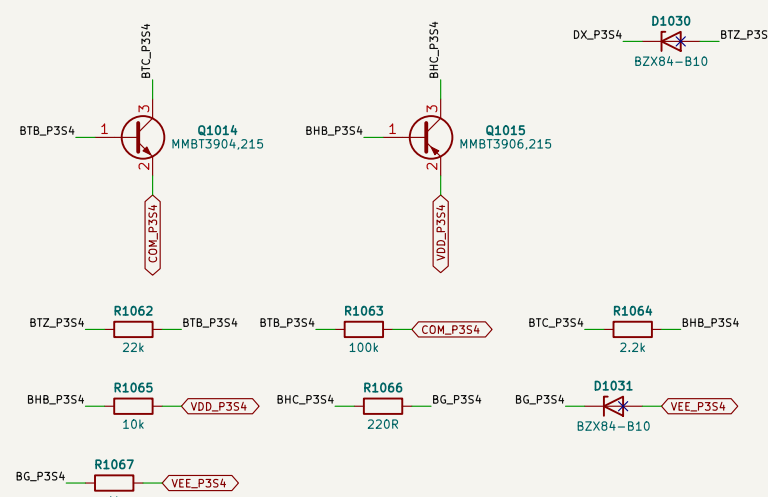
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



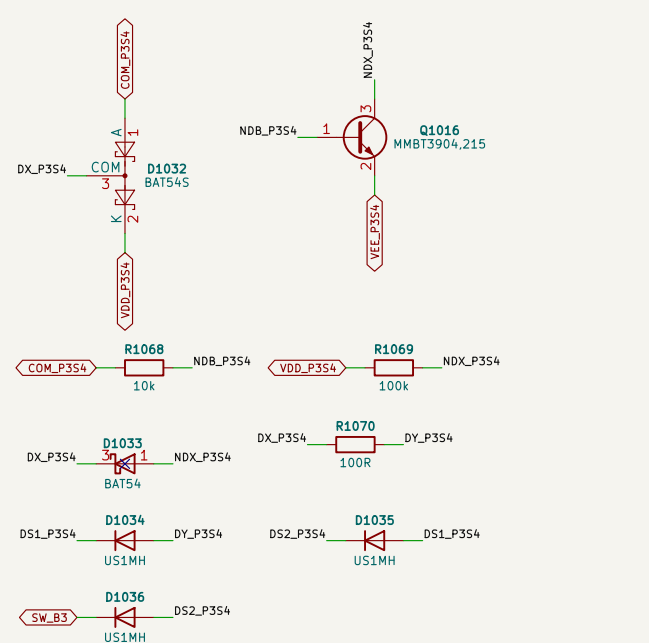
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nigpice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5. C = BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array I_{sc} ≤ 1.25 x rating.
 Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T, Up,eff 3793 V pole-P and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); Imax 25 kA, Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Not bench-validated. Uses read from sim/out/*.json; checks: design_check() Sheet 12		
Schematics authored by Chinmoy Bhuyan Live board referenced to BUS--; gate islands insulating; SELV winding reinforced; HV-PE basis		
Sheet: /11 Port A power path (pv)/ File: 11_leanA_power.kicad_sch		
Title: PV-PWR power board – Port A power path (pv)		
Size: A2 KiCad E.D.A. 10.0.6	Date: 2026-10-05	Rev: A2 Id: 12/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /12 Port A sensing and interlocks/
File: 12_leanA_ctrl.kicad_sch

Title: **PV-PWR power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 13/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 14 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /13 Port A coil drives/ File: 13_leanA_drive.kicad_sch		
Title: PV-PWR power board – Port A coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 14/23	

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

[illegible]

TV225751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE-1, U_{eff} 3793 V pole-PE and pole-pole at 3 kA (calc. = 4.0 kV) => reinforced 6 kV / basic 4 kV, D=032; I_{max} 25 kA, Branch fuses E1 (SMT) 50kV 35 A (30 kA / I₂ t₂ ms), Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

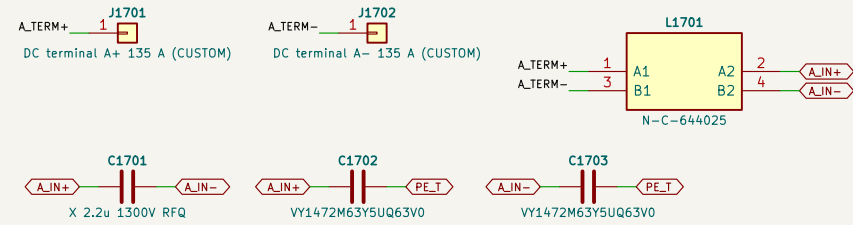
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 16 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /14 Port B sensing and interlocks/ File: 14_leanB_ctrl.kicad_sch		
Title: PV-PWR power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 16/23	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

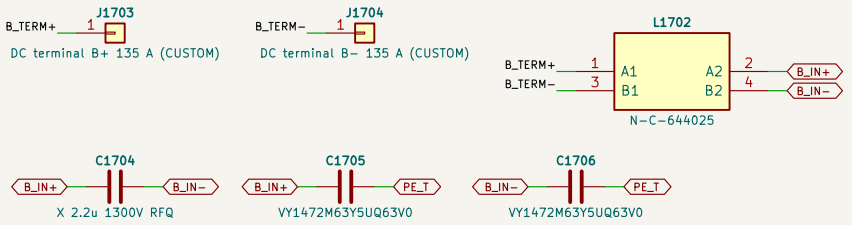
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 17 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port B coil drives/ File: 15_leanB_drive.kicad_sch		
Title: PV-PWR power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 17/23	

PORT A TERMINALS AND EMI FILTER



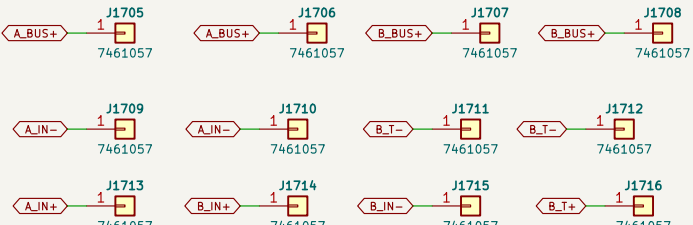
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



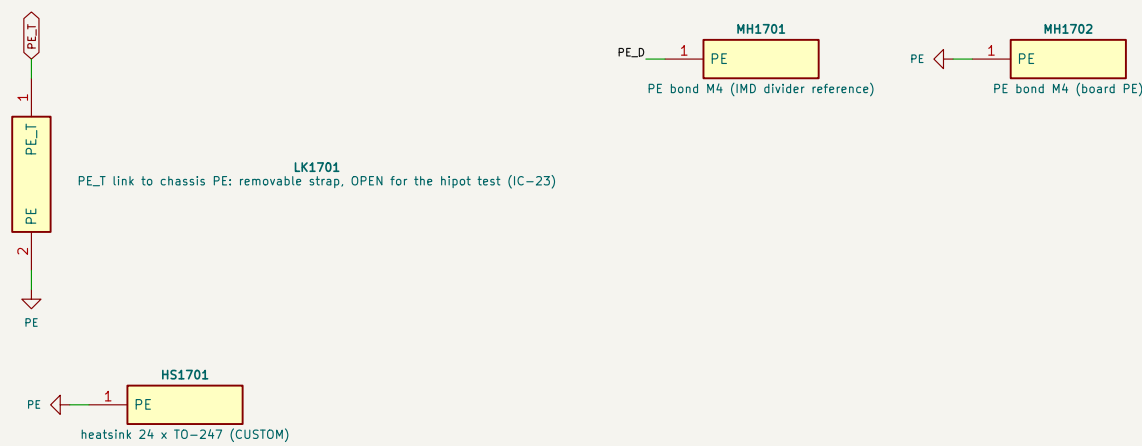
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



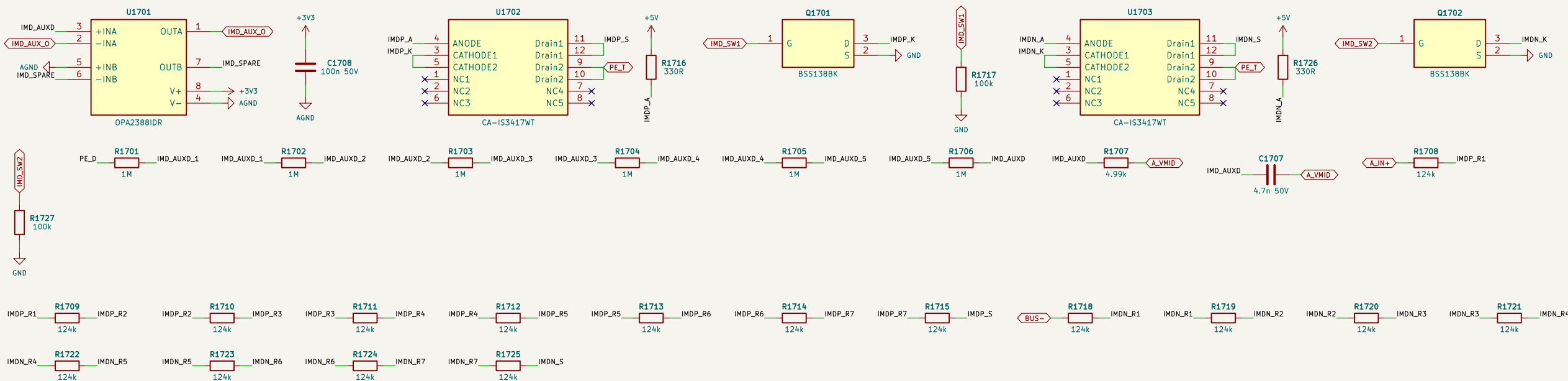
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected);
VPE = PE divider against BUS- (VMID of port A).

Not bench-validated: Values read from sim/out/*.json; checks: design_check()

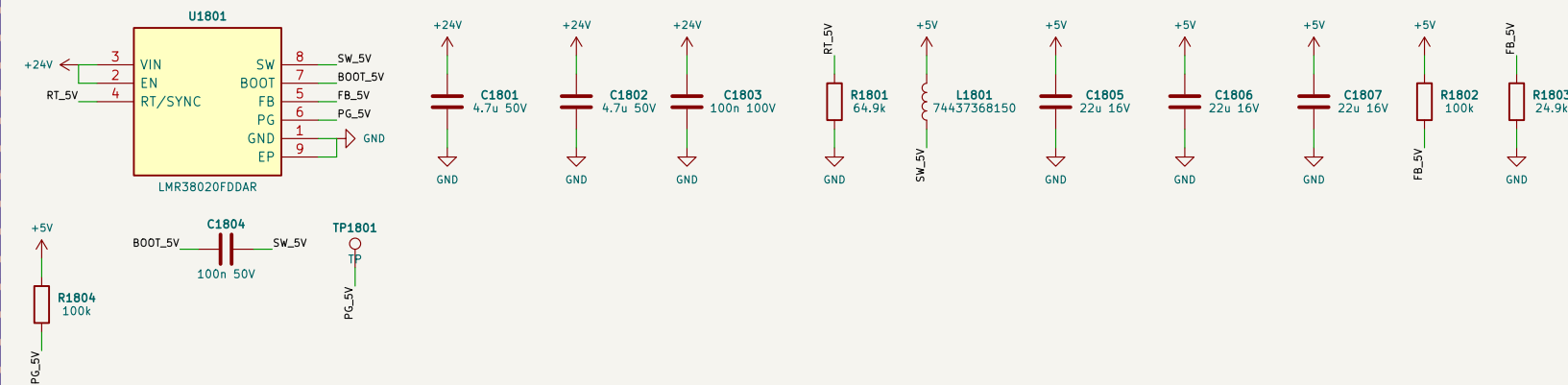
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Terminals, EMI filter, insulation test/
File: 15_terminals_imd.kicad_sch

Title: PV-PWR power board - Terminals, EMI filter, insulation test

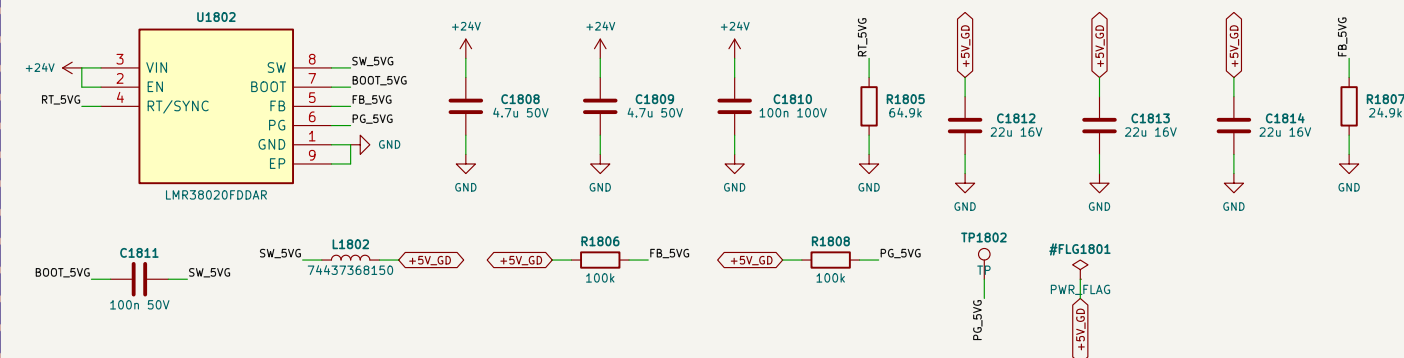
Size: A3 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 18/23

24 V $\rightarrow$ +5V (LMR38020, 400 KHZ, 2 A)



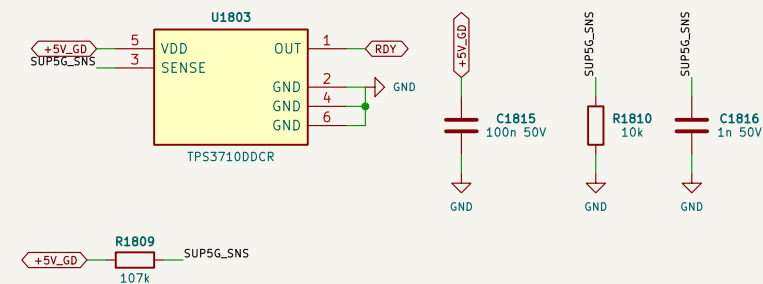
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2. ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



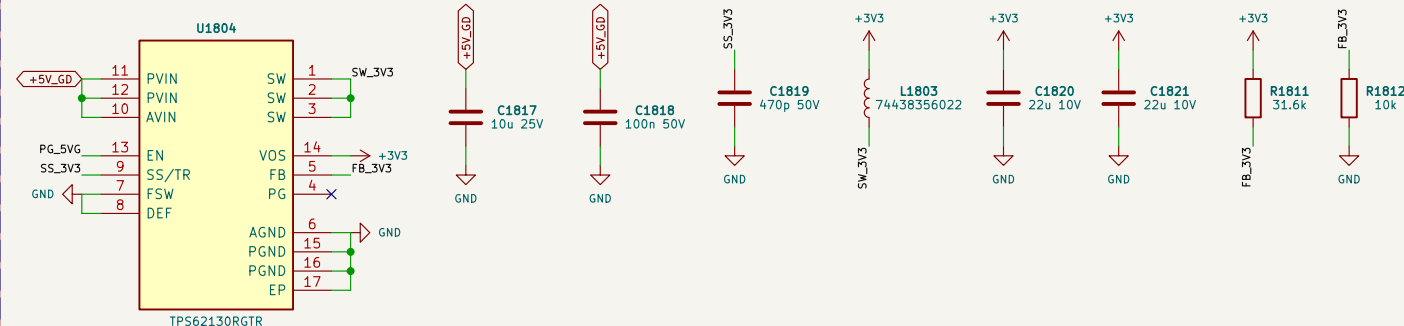
gate bias of phases 3 only. ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



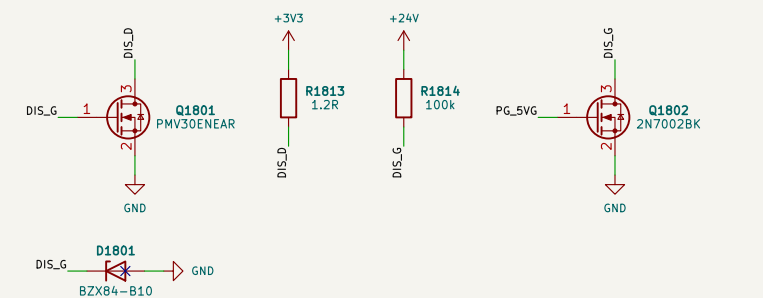
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52–4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5V0. FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (≥ 3.20 V for the control board's reference). SS/TR 470p COG: ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control ≤ 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENE4 + 1.2R: +3V3 falls at ≥ 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 21.4–30.1 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 19

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Live 5 V / 3.3 V, SELV connector/

File: 16_supplies.kicad_sch

Title: PV-PWR power board – Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 19/23

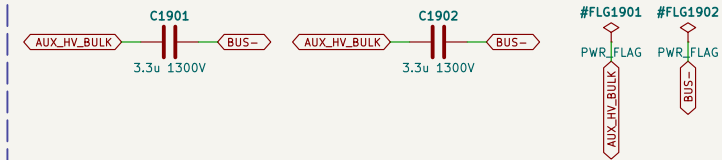


2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

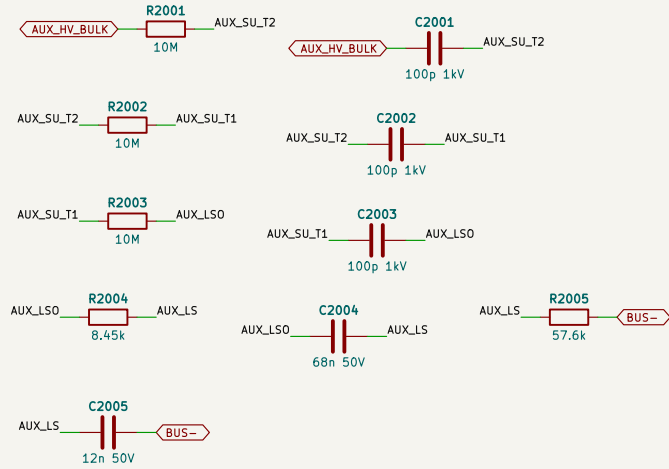
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 AUX75 input taps, OR, film/
File: 17_aux75_input.kicad_sch

Title: PV-PWR power board – AUX75 input taps, OR, film

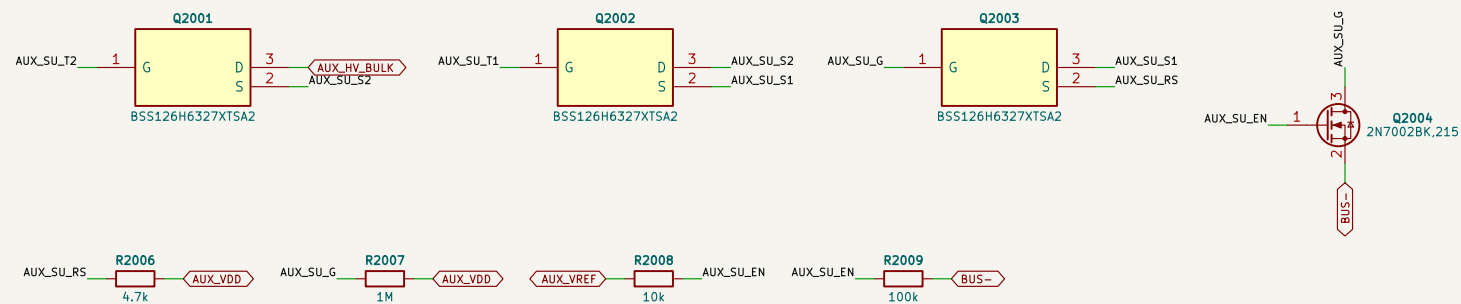
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 20/23

COMPENSATED TAP STRING



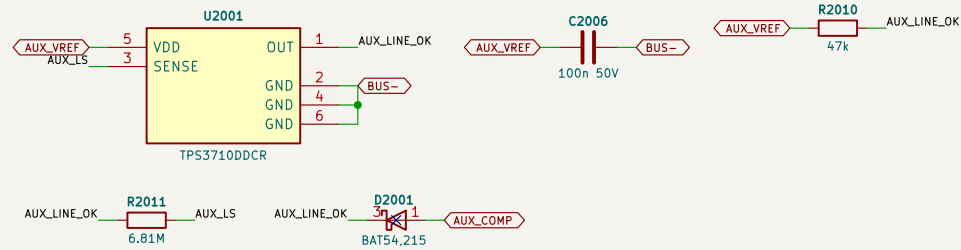
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



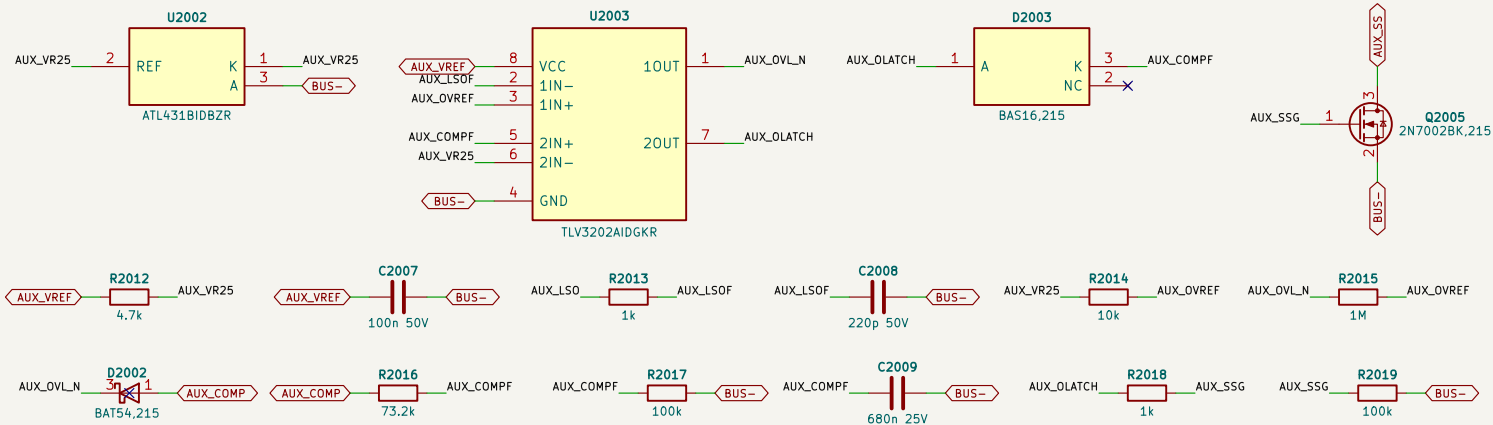
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



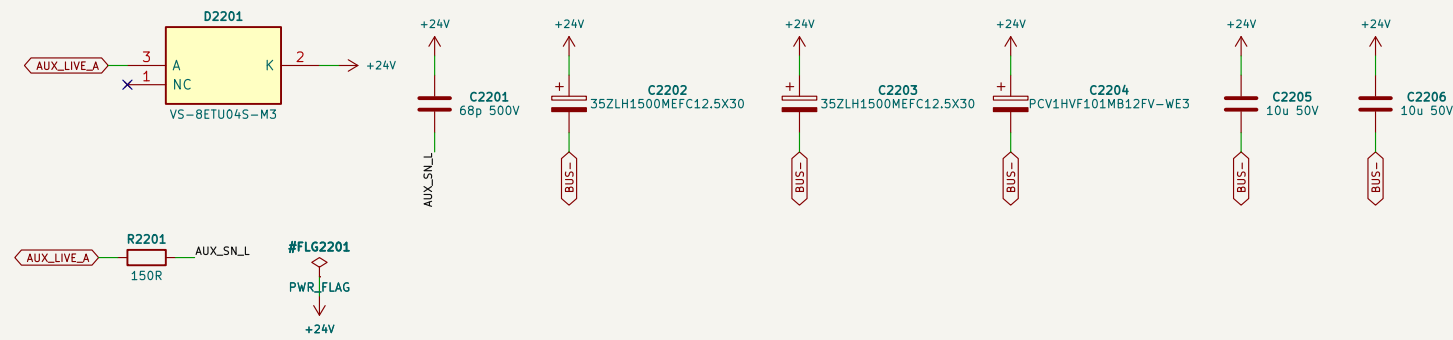
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



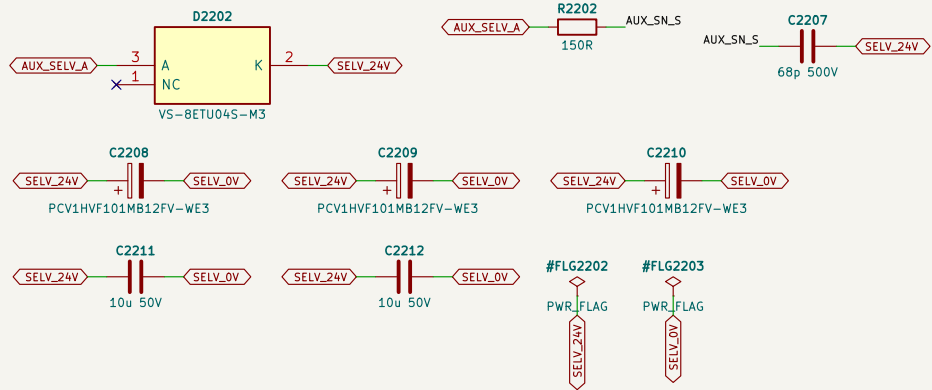
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms -> latch -> SS crowbar -> hiccup.

LIVE 24 V RECTIFIER, HOLD-UP



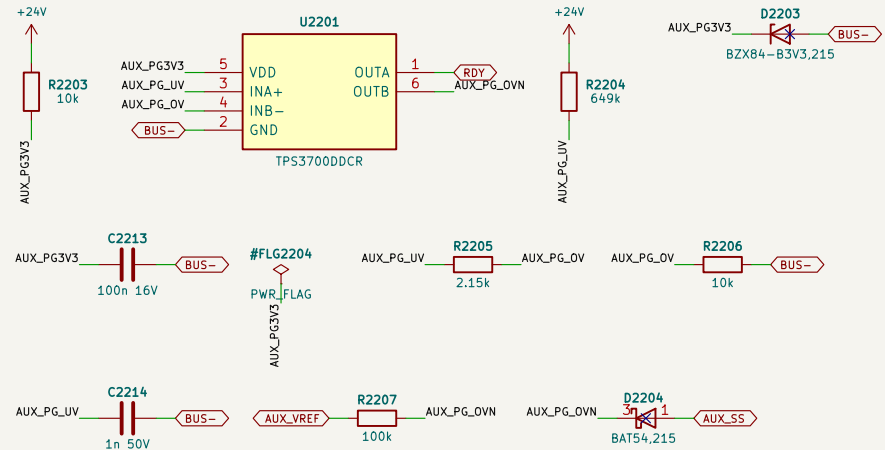
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)